

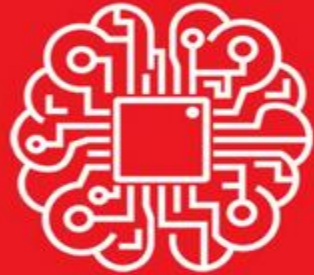
Artificial Intelligence as a driver for Innovation in Official Statistics

Francesco Pugliese, Central Directorate for Methodology and Design of Statistical Processes, **Istat**

ARTIFICIAL INTELLIGENCE AND OFFICIAL STATISTICS: THE NEW FRONTIER OF INNOVATION

Napoli | 09 April 2026 | Istat of Campania

Difference between AI and ML



ARTIFICIAL
INTELLIGENCE

If it is written
in **PowerPoint**,
It's probably
**Artificial
Intelligence**



MACHINE
LEARNING

If it is written
in **Python**;
It's probably
**Machine
Learning**

MACHINE LEARNING ENGINEER



What My Parent Think I do



What My Friend Think I do



What Society Think I do



What Media Think I do



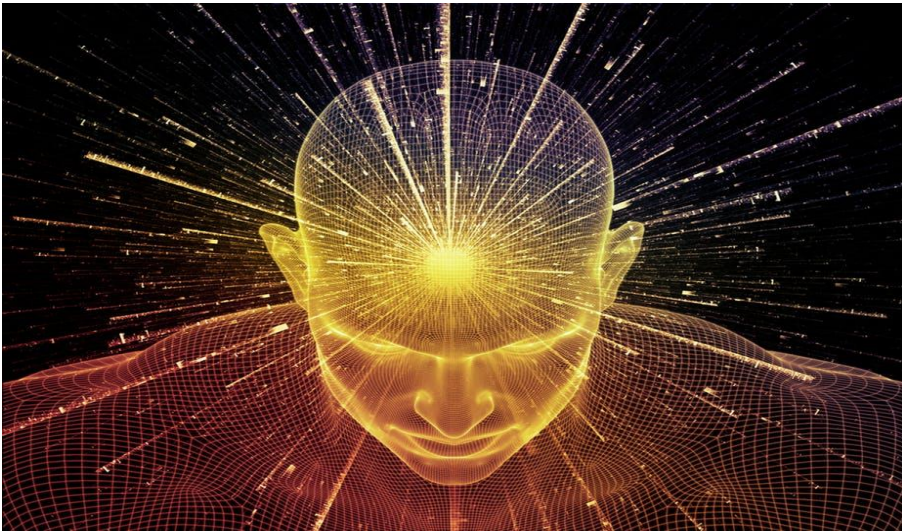
What I Think I do

```
1 import tensorflow as tf
2 import torch
3
```

What I really do

WHAT IS CONSCIOUSNESS?

WHAT IS THE SUBJECTIVE EXPERIENCE? NOBODY KNOWS



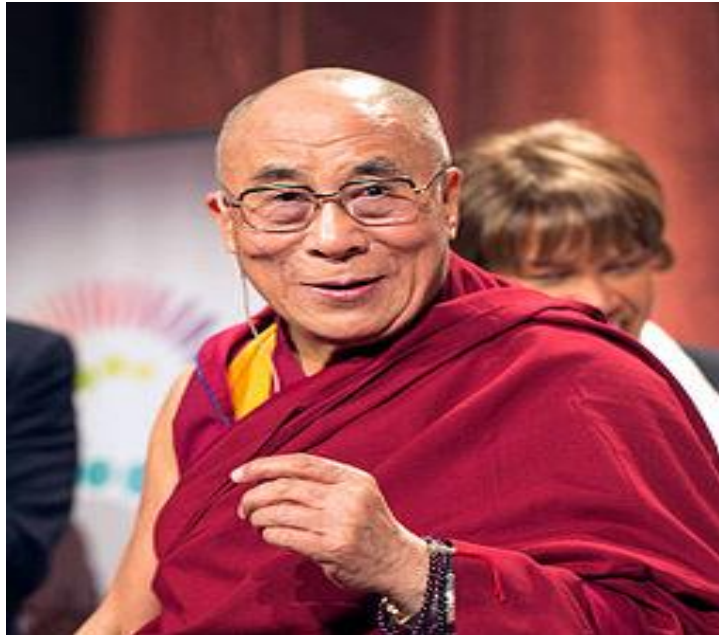
DISSOCIATION BETWEEN CORTICAL ACTIVITY AND COUSCIOUSNESS

- During episodes of unconsciousness, the child's brain is by no means inactive; it is actually hyperactive.
- The observation of similar firing rates between waking and slow-wave sleep represents both the most striking dissociation between cortical activity and consciousness.
- It indicates that the presence of “normal” firing levels in the cerebral cortex is not a sufficient criterion for consciousness.

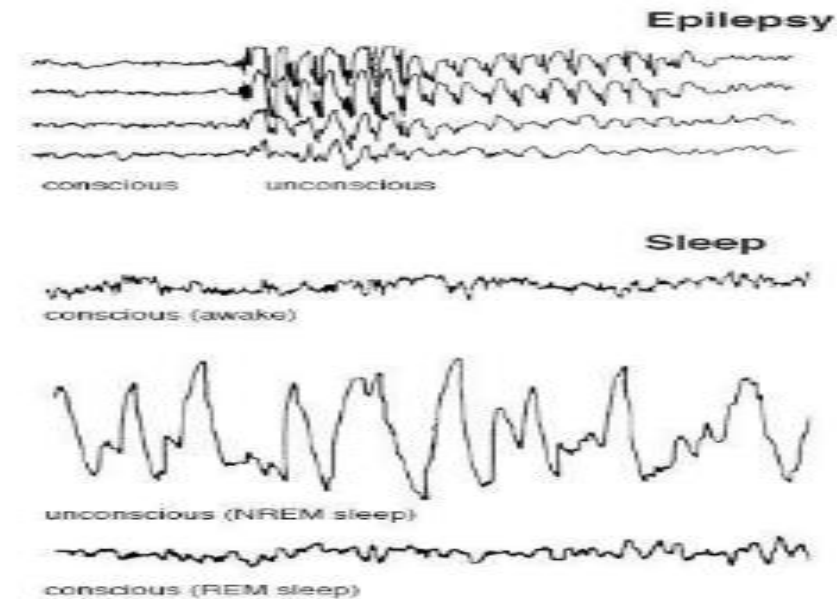


Edelman, Gerald. A Universe Of Consciousness: How
Matter Becomes Imagination

How Thoughts can change the Brain



Conscious Experience requires PATTERNS OF NEURAL ACTIVITY THAT ARE HIGHLY DIFFERENTIATED (Gerald Edelman)



Tenzin Gyatso (Dalai Lama): “What if the brain comes from consciousness instead of consciousness coming from the brain?”

HUMAN BRAIN VS CONSCIOUSNESS

An High Dimensional and Complex Neural Network



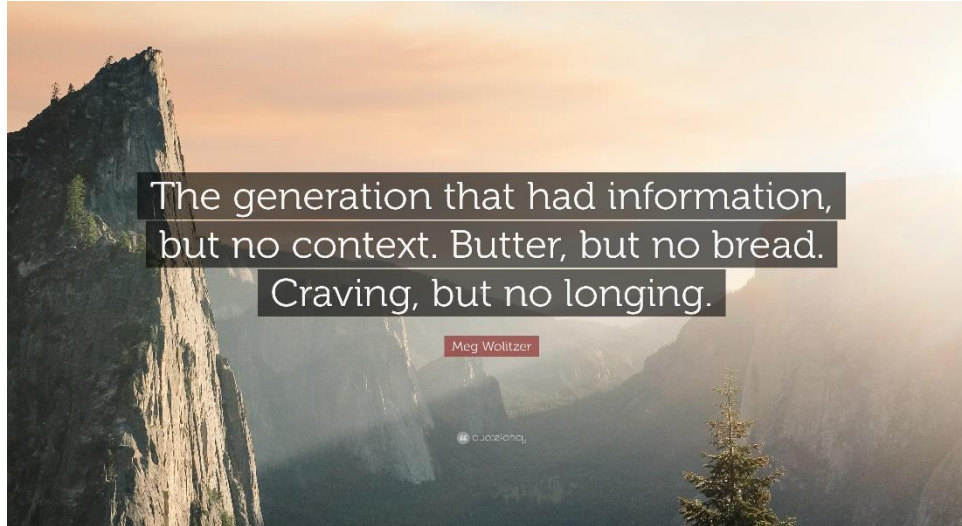
- The lack of a formal link between neural network structure and its emergent function has hampered our understanding of how the brain processes information.

Scientists find evidence of a Multidimensional Universe within the Brain



Reimann, M.W., Nolte, M., Scolamiero, M., Turner, K., Perin, R., Chindemi, G., .. & Markram, H. (2017). Cliques of neurons bound into cavities provide a missing link between structure and function. *Frontiers in computational neuroscience*, 11, 48.

MACHINE LEARNING VS CONSCIOUSNESS



Is this a Glass or a Pot?



Where is the Human-Like Thoughts Flow?

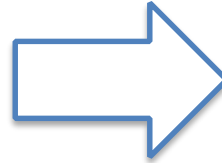
**Why and How Lea Seadol beats Alpha
Go Zero
once?**

**Even though all metrics say it is
impossible.**

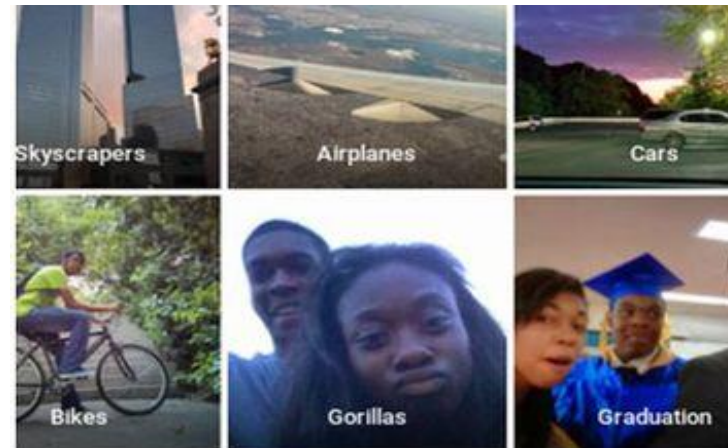


MACHINE LEARNING LIMITS

Will ever Self Driving Car cope with Mumbai Traffic?

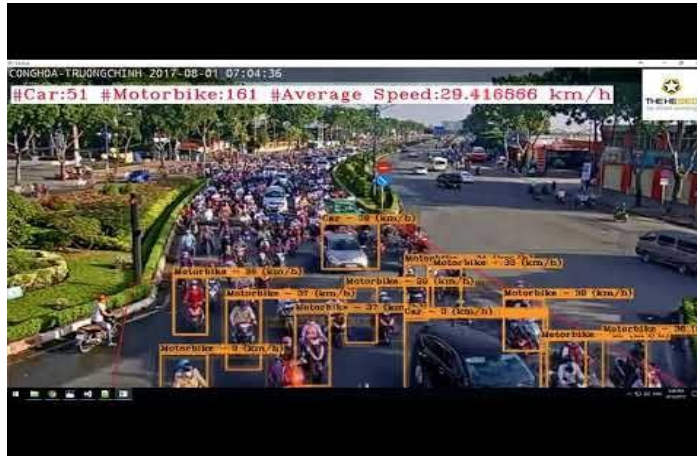


Will ever AI have context information ?

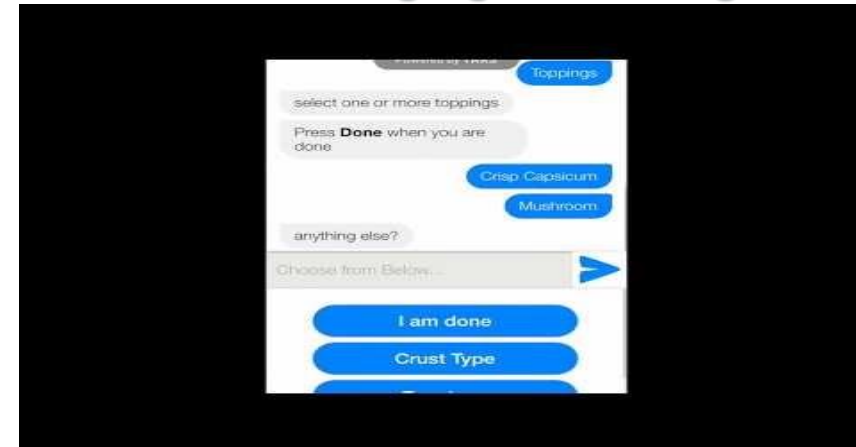


MACHINE LEARNING BREAKTHROUGHTS

Computer Vision



Natural Language Processing



A truly general Artificial Intelligence

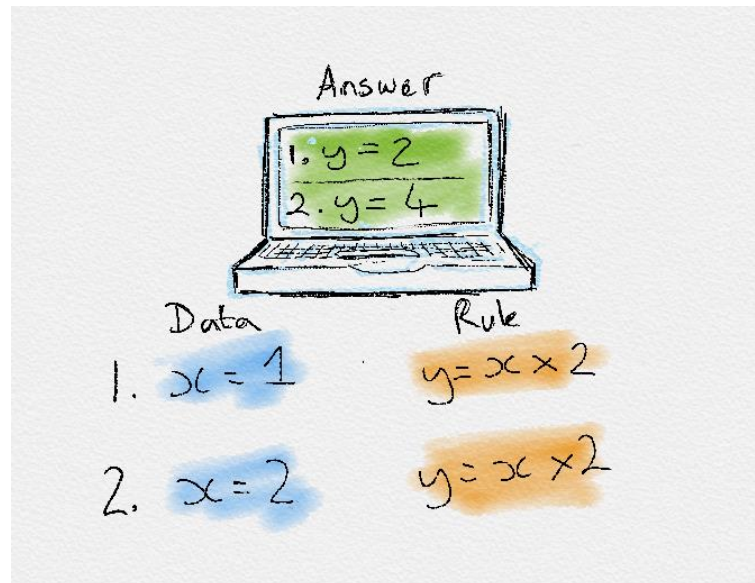


A Generative Artificial Intelligence

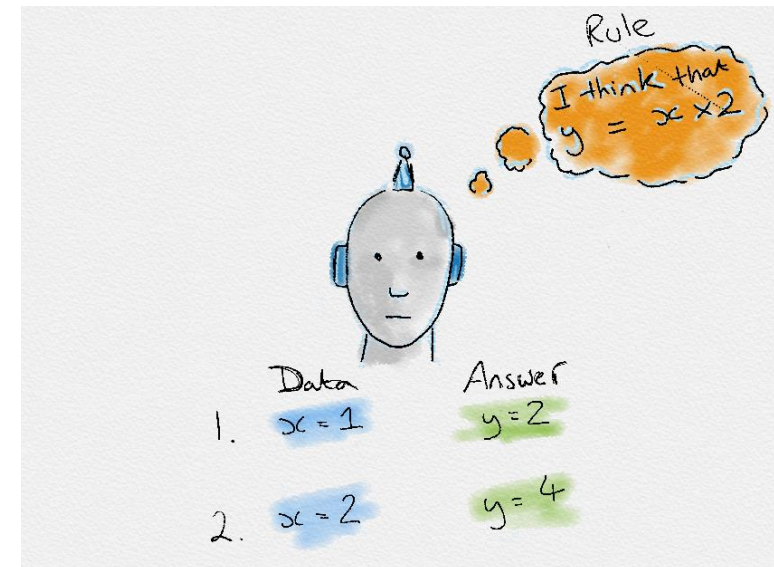


Traditional programming vs Machine Learning

- ✓ Machine learning is a subfield of computer science that is concerned with building algorithms which, to be useful, rely on a collection of examples of some phenomenon.
- ✓ Traditionally, software engineering combined human created rules with data to create answers to a problem. Instead, machine learning uses data and answers to discover the rules behind a problem.
- ✓ To learn the rules governing a phenomenon, machines have to go through a learning process, trying different rules and learning from how well they perform. Hence, why it's known as Machine Learning.

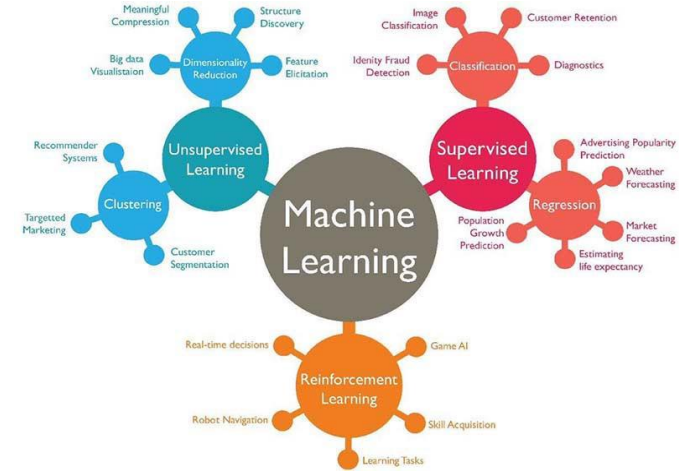


VS

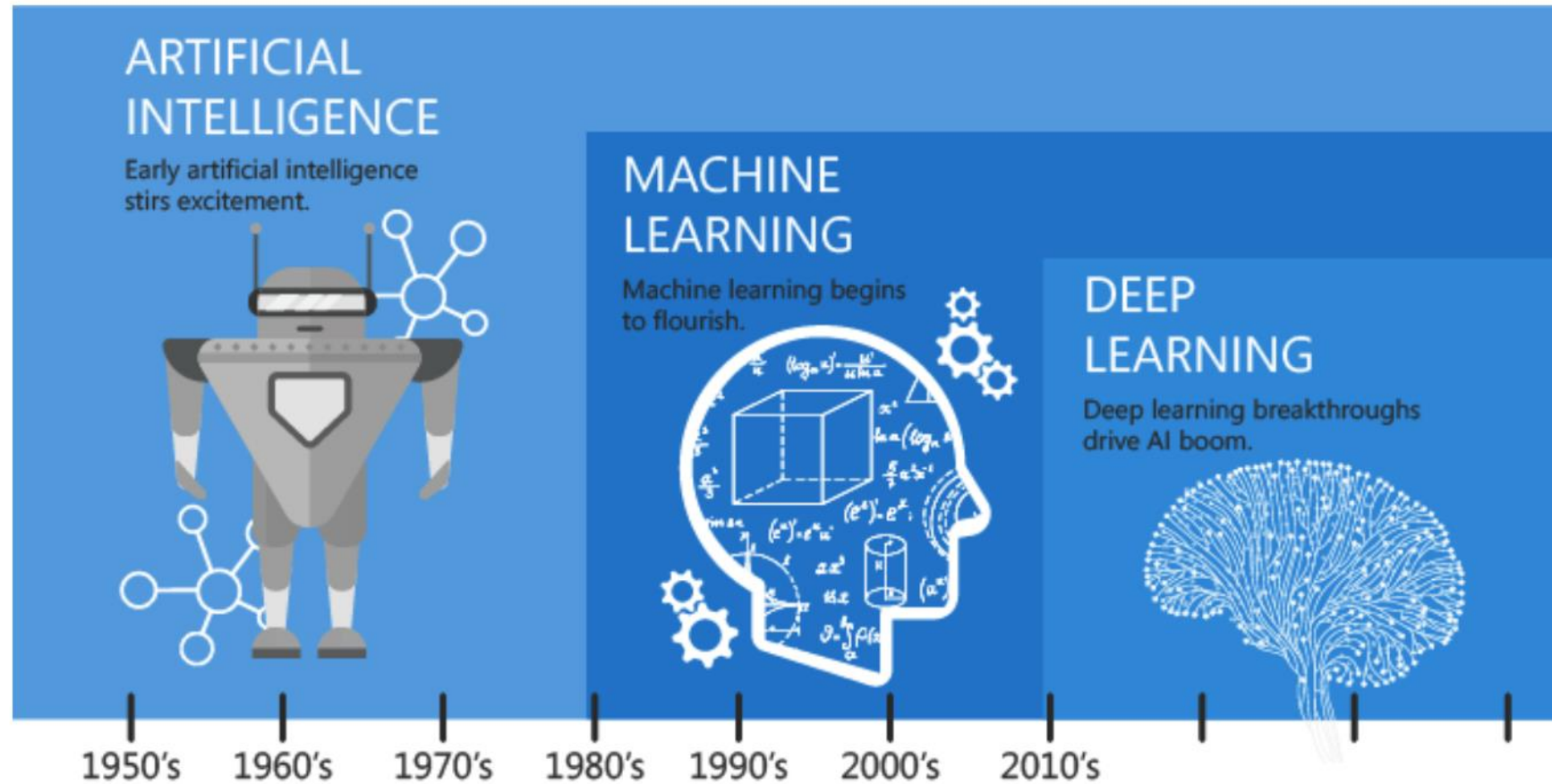


Types of Machine Learning

- ✓ There are broadly 4 types of Machine Learning namely,
 - ✓ Supervised learning
 - ✓ Unsupervised learning
 - ✓ Semi-supervised learning
 - ✓ Self-supervised learning
 - ✓ Reinforcement learning
- ✓ Each form of Machine Learning has differing approaches, but they all follow the same underlying process and theory.
- ✓ The **No Free Lunch theorem** is famous in Machine Learning. This theorem states that there is no single algorithm that works well for all types of tasks. Each task you attempt to solve has its own idiosyncrasies, meaning its own specificities and adversities to certain Machine Learning models. Therefore, there are many algorithms and approaches that adapt to each specificity and peculiarity of an individual problem.



A quick history of Machine Learning

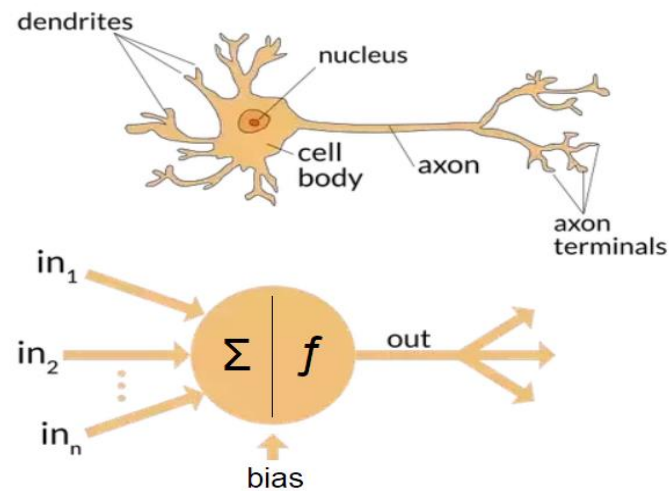


Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

ARTIFICIAL NEURAL NETWORKS

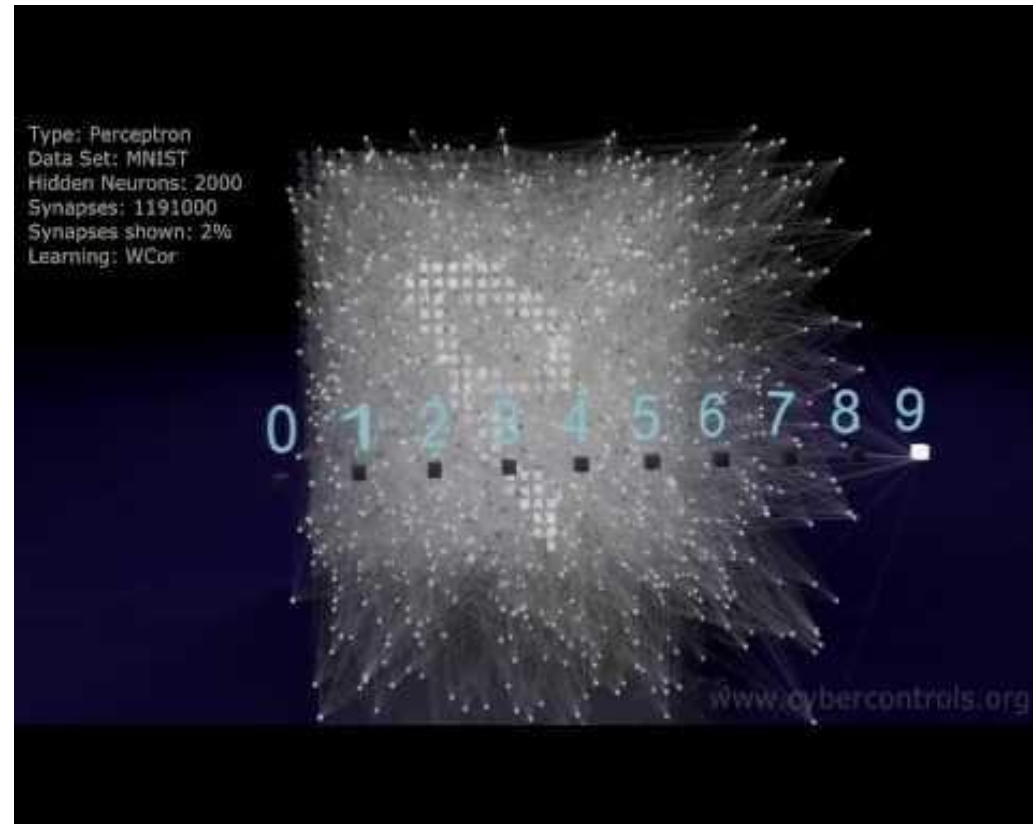
- **Artificial neural networks (ANN) or Connectionist Systems** are computing systems vaguely inspired by the biological neural networks that constitute animal brains (van Gerven & Bohte, 2018)

Biological Neuron Vs Artificial Neuron



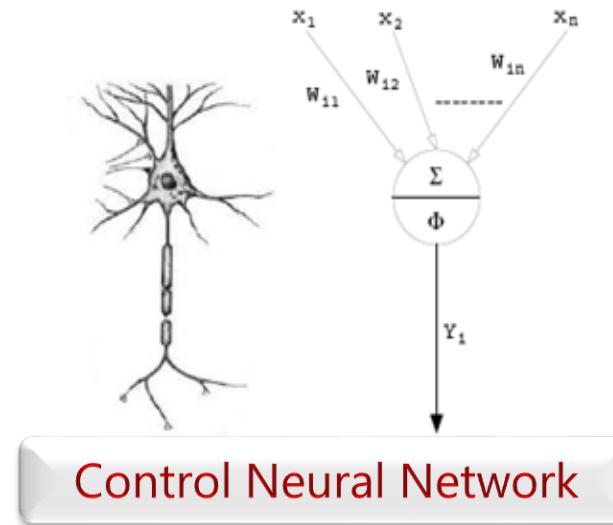
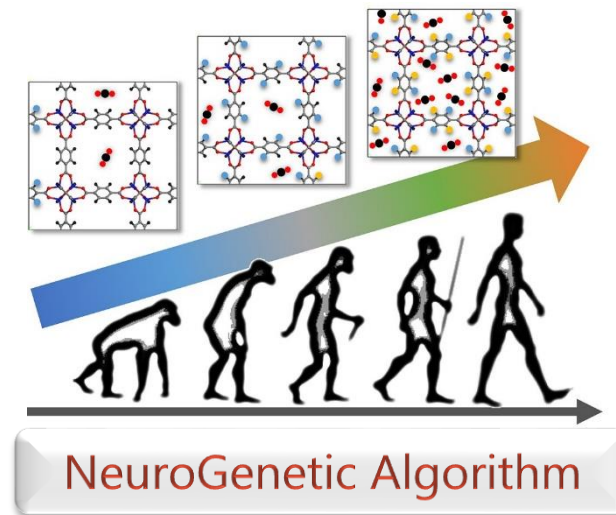
Neuron Activation

$$y_j = \Phi\left(\sum_{i=1}^N w_{ij}X_i - \theta_i\right)$$

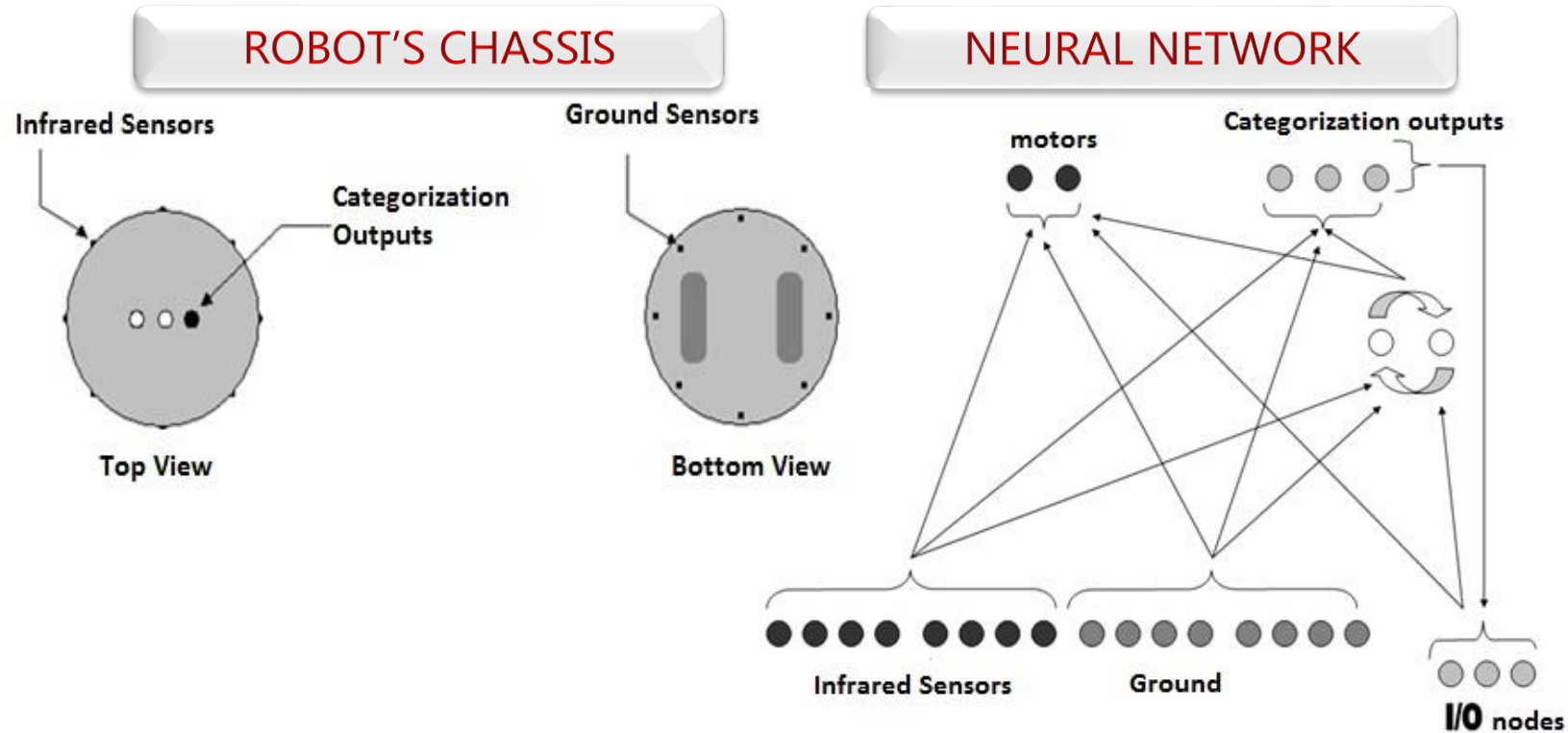


One traditional application ANN: Evolutionary robotics

IN ORDER TO OVERCOME THE PROBLEMS ASSOCIATED WITH THE ROBOTIC SYSTEM DECOMPOSITION OF TRADITIONAL APPROACHES (I.E. BEHAVIOR-BASED ROBOTICS), EVOLUTIONARY ROBOTICS CAN BE USED, WHERE THE ROBOTIC SYSTEM IS ABLE TO SELF-ORGANIZE
[NOLFI, S., FLOREANO, D., 2002].



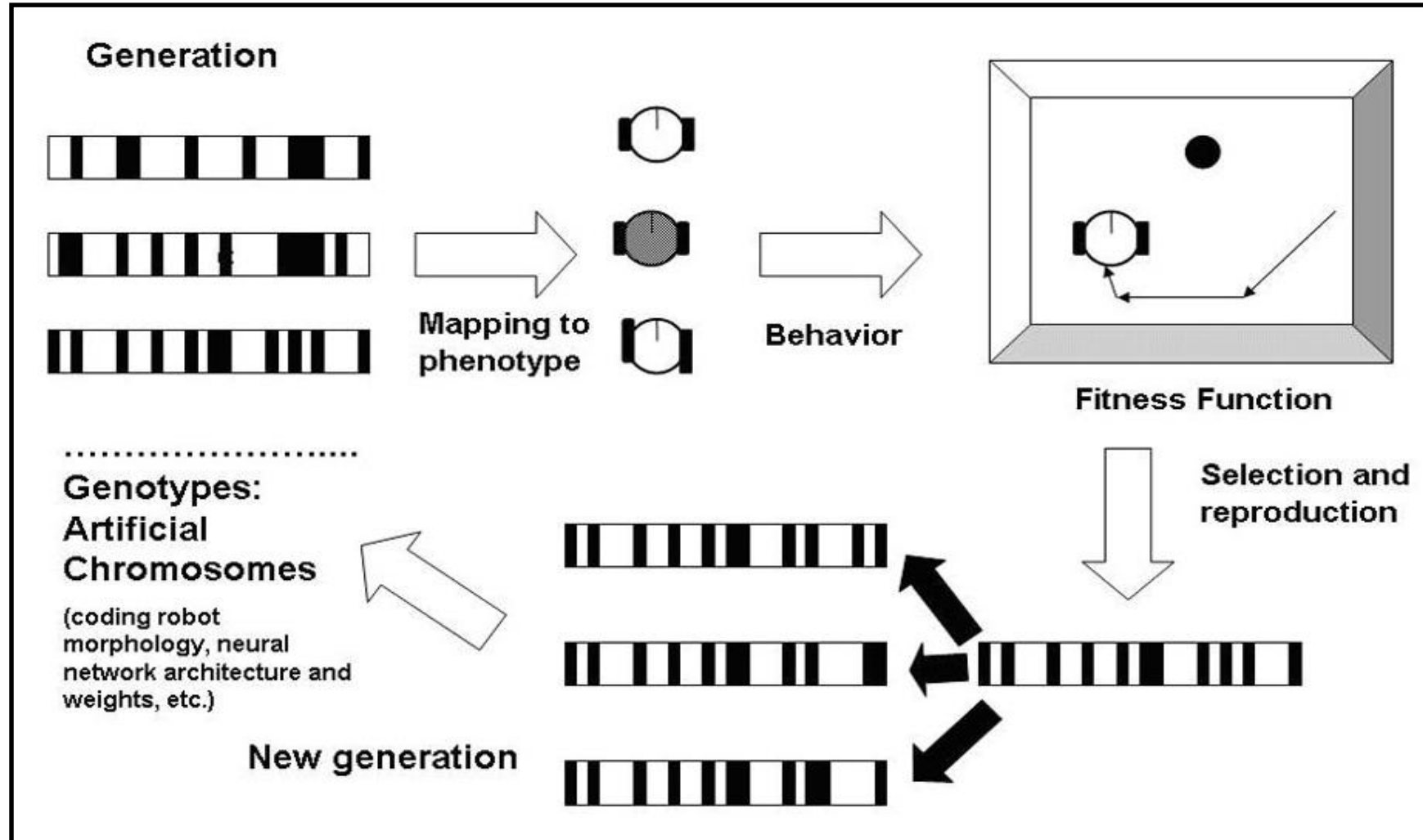
EXPERIMENTAL SETUP



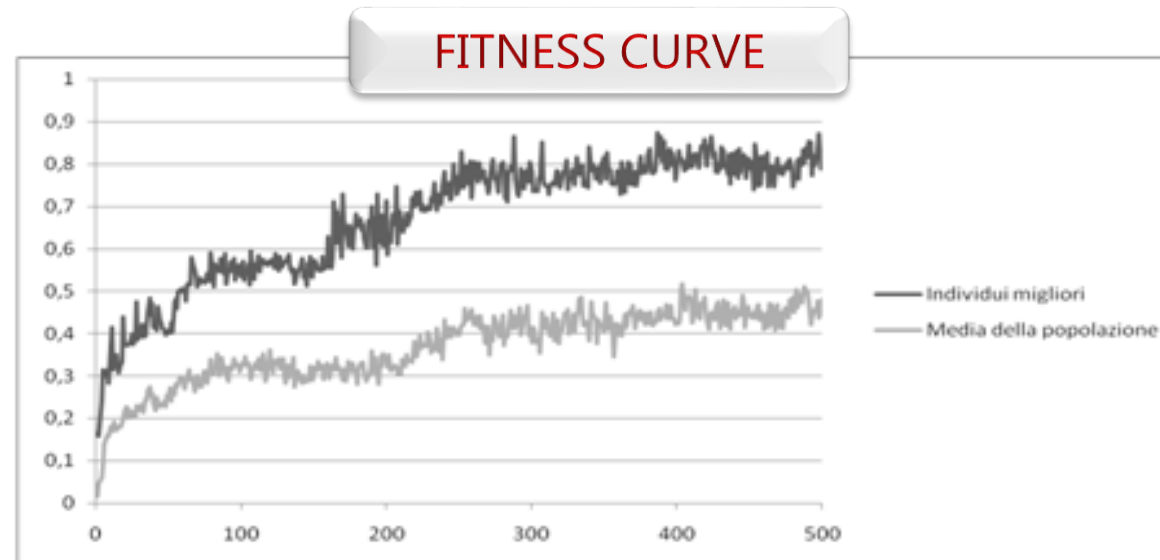
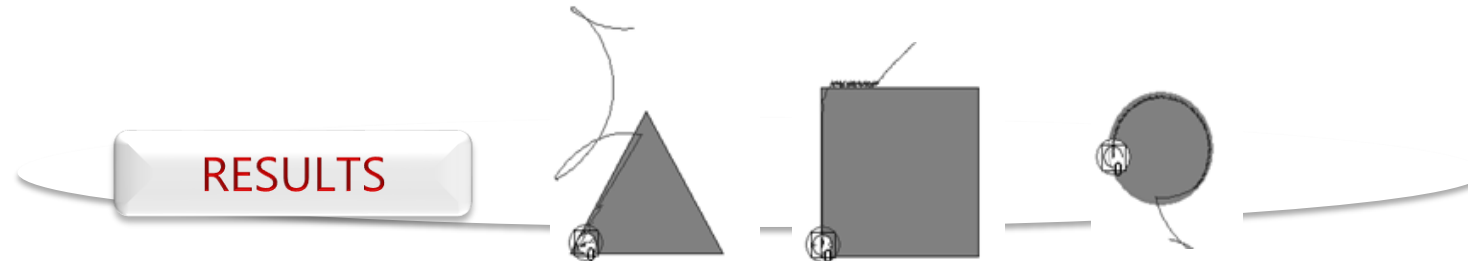
LEAKY Activation

$$A_j = t_j + \sum w_{ij} O_i, O_j = \delta_j O^{t-1} + (1 - \delta_j) \left(1 + \frac{1}{e^{A_j}} \right), 0 \leq \delta_j \leq 1$$

Evolutionary robotics



EXPERIMENTAL SETUP N.1

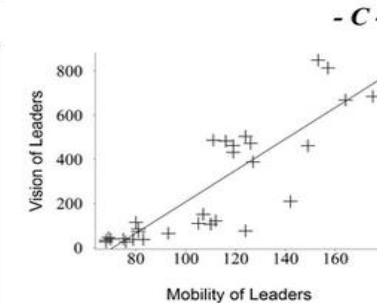
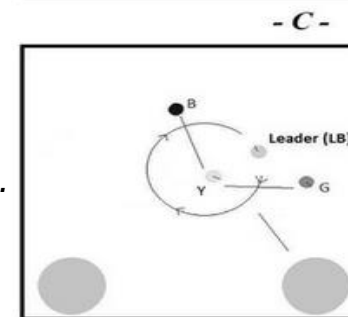
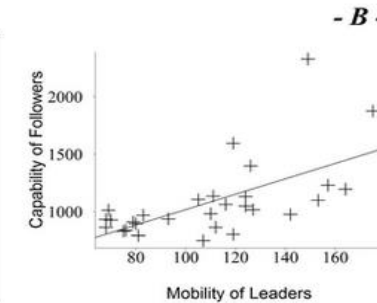
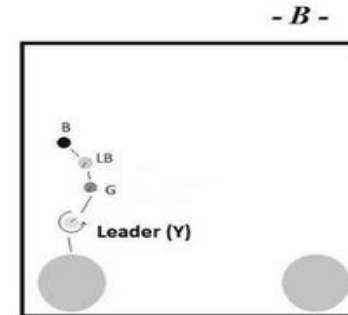
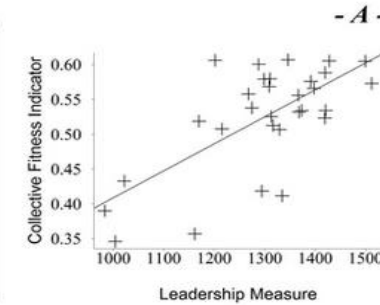
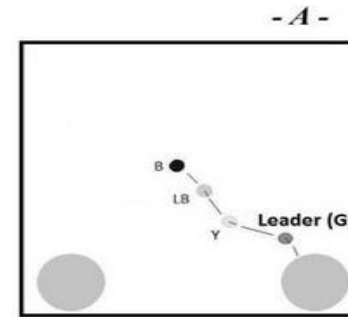


Emergence of Leadership in Robots

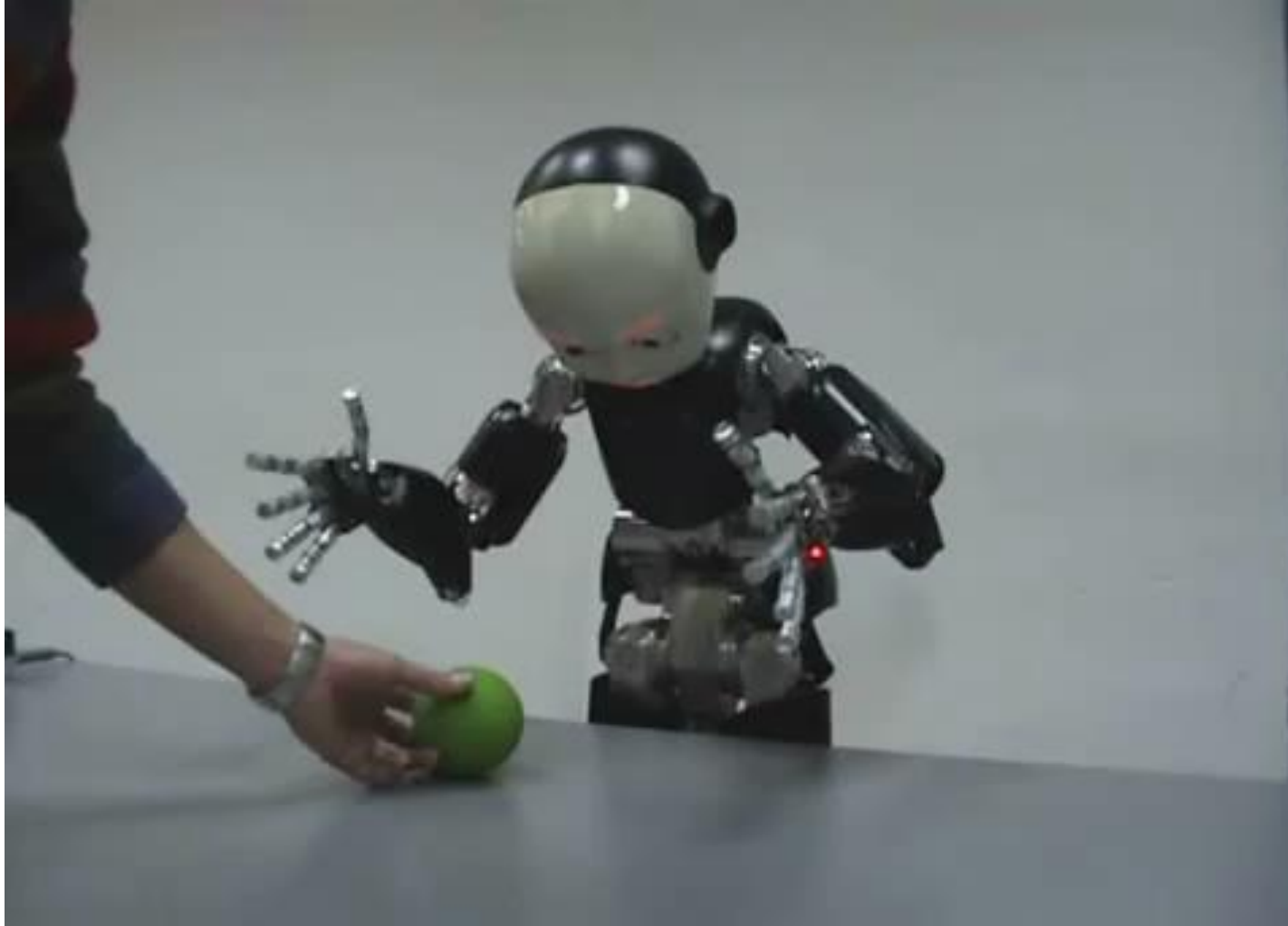
- Behavioural and quantitative analysis indicate that a form of leadership emerges
- Groups with a leader are more effective than groups without.
- The most skilled individuals in a group tend to be the leaders.
- Further analysis reveals the emergence of different “styles” of leadership (active and passive).

- A - *Passive Leadership.* - B - *Weak Active Leadership.*
- C - *Strong Active Leadership.*

- (Pugliese, et. Al., 2015)



MAY Robotics help to understand social and psychological problems?



DEEP LEARNING: Neural Networks become more effective

In recent years **Deep Neural Networks** have achieved noticeably breakthroughs in research (Bengio, 2009). This new methodology dealing with deep neural networks and their training algorithms is called “*Deep Learning*”. So far, in all the experiments, the resulting performances were many magnitudes better than other machine learning techniques available.



GOOGLE DATACENTER

1,000 CPU Servers
2,000 CPUs • 16,000 cores

600 kWatts
\$5,000,000

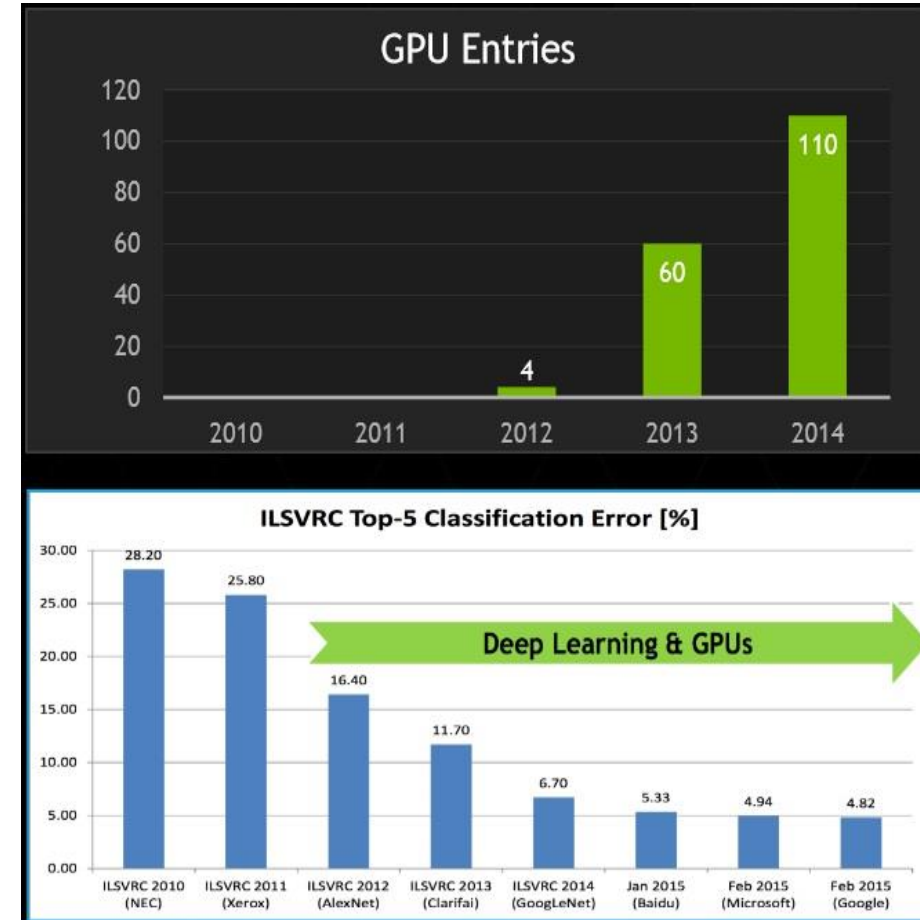
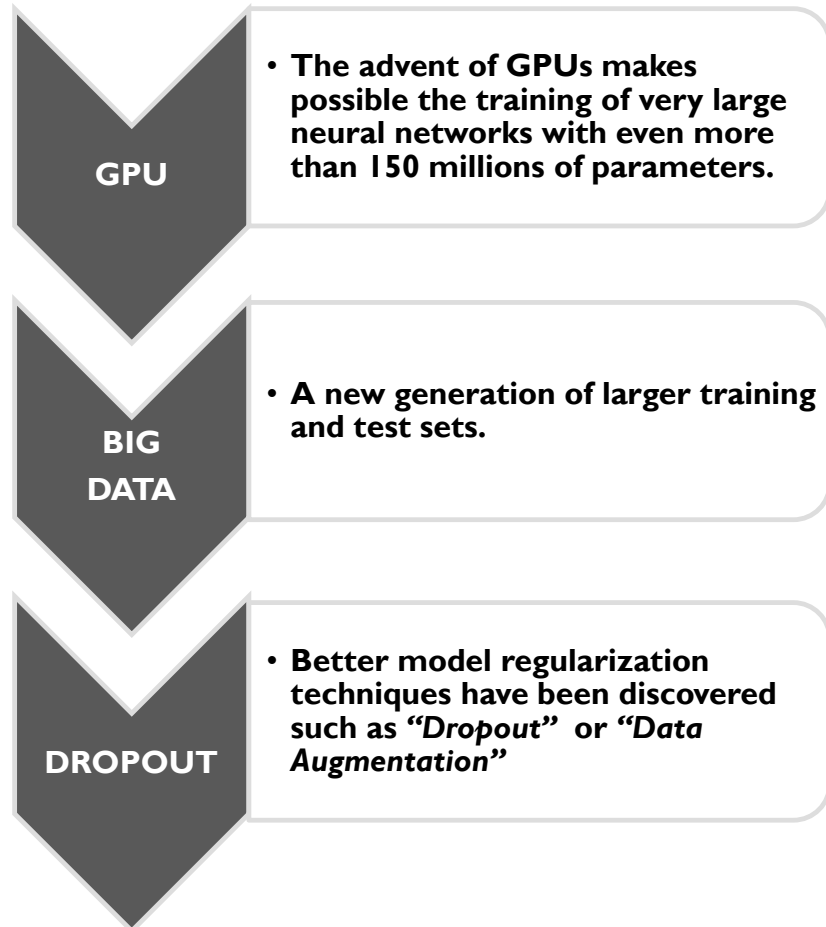


STANFORD AI LAB

3 GPU-Accelerated Servers
12 GPUs • 18,432 cores

4 kWatts
\$33,000

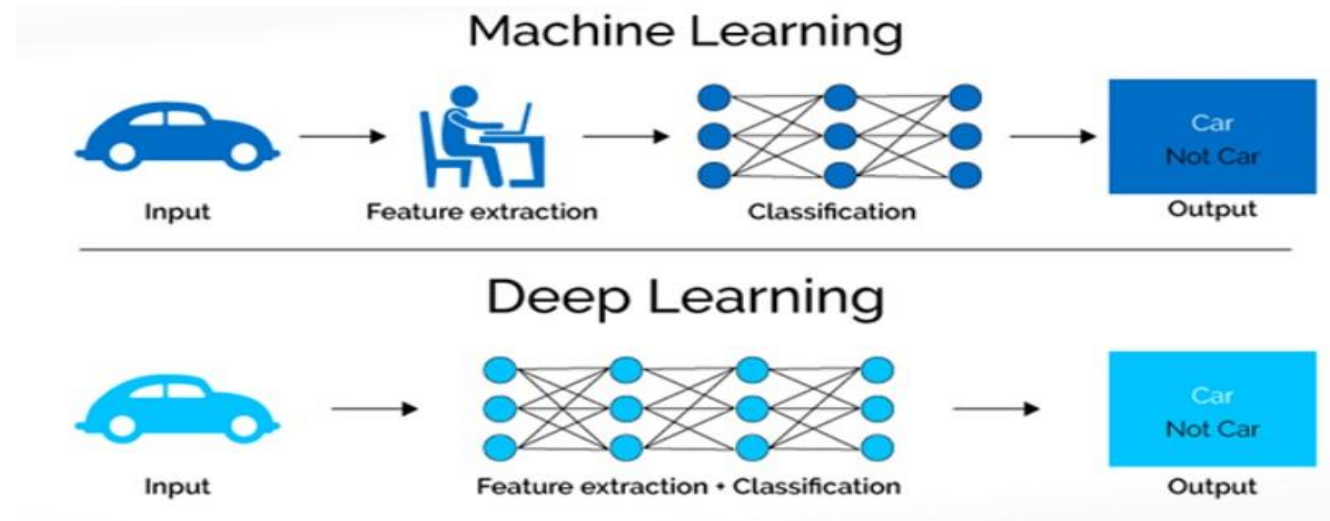
DEEP LEARNING: a cutting-edge approach to Computer Vision and NLP



Deep Learning is Machine Learning

Deep Learning refers to algorithms that automatically 'model' high-level abstractions in data

- i. here 'model' means: define, find, recognize and exploit
- ii. here 'automatically' means: directly from data, without hinging upon handcrafted, task-specific features.



Why Deep Learning over-performed traditional statistics models?



- “Deep Learning” approaches can be **end-to-end trained** without a task-specific feature engineering.
- **These models are scalable:** adding GPUs they can be trained faster.
- “*Deep Learning is killing every problem in AI*” (Elizabeth Gibney, 2016)
- Basically, **statistics is not able to deal with very high dimensionalities** of data as Deep Learning does.

23

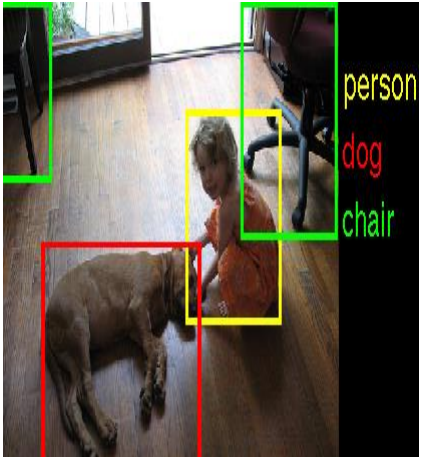
Computer Vision: Where does Traditional Statistics fail?



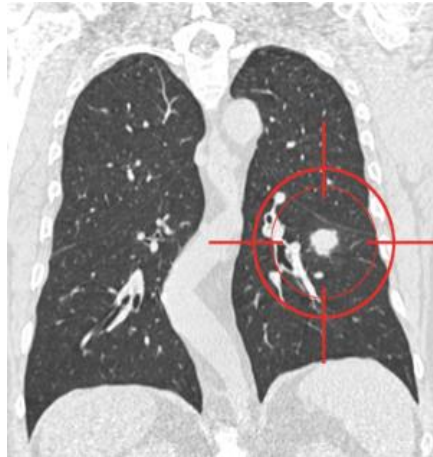
- **Computer Vision** is an interdisciplinary field that deals with the way algorithms can be made for gaining high-level understanding from digital images or videos.
- **Statistical methods** are not always welcome in computer vision.
- Statistical methods seem **not scaling up** to the challenges of computer vision problems (Chellappa, R., 2012).

Why does Computer Vision matter so much?

EVERYDAY LIFE



BIOMEDICAL IMAGES



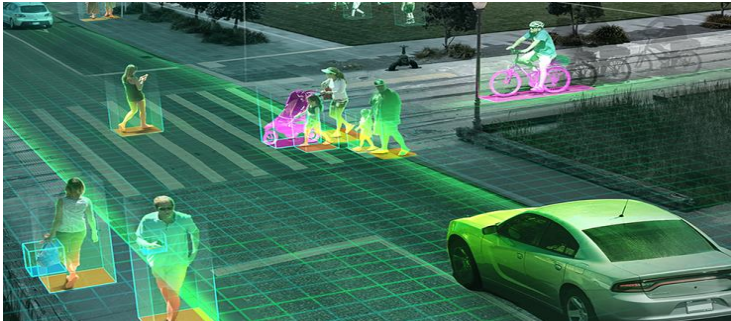
- A new study proves the **relationship** between **Vision capabilities and Intelligence** (Tsukahara et al., 2016).
- Seemingly, **human beings** have one of **the most complex vision systems** within the **animal realm** and this fact would have fostered **the evolution of human intelligence**.
- In other words, **Computer Vision** needs **human-like intellectual capabilities** in order to achieve the vision performances required by humans for every-day applications.

25

Why does Computer Vision matter so much?



- **A new generation of machines** might accomplish typical human tasks such as recognizing and moving objects, driving cars, cultivating fields, cleaning streets, city garbage collecting, etc.



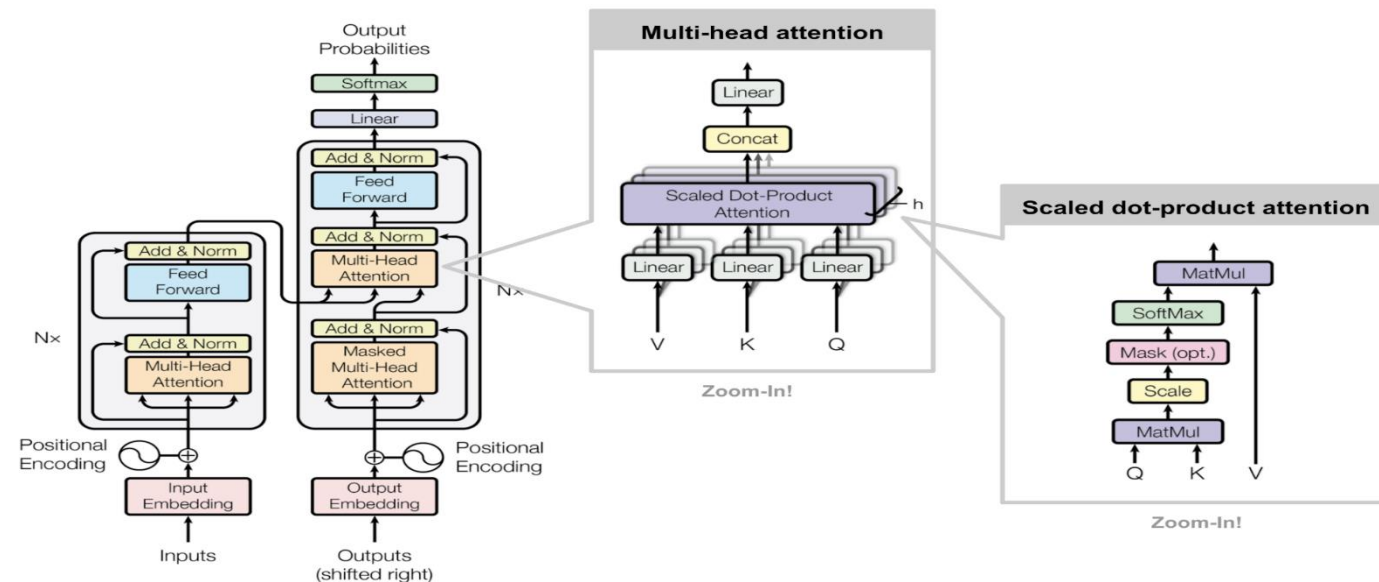
The Innovation of Transformers

- ✓ **Transformers** change all the game in **Natural Language Processing (NLP)**, which has become one of the pillars of artificial intelligence in a global digital economy.
- ✓ **Transformer** models are disruptive and are leading to a new age in **Artificial Intelligence**. Language understanding has become the mile-stone of language modeling, chatbots, personal assistants, question answering, text summarizing, speech-to-text, sentiment analysis, machine translation, and more.
- ✓ By means of **Transformers**, **Artificial Intelligence** has evolved to billion-parameter models in order to address the challenge of trillion-word datasets.
- ✓ They get far from the past, leaving the dominance of **RNNs** and **CNNs** behind. **BERT** and **GPT** models leave the recurrent network layers back and replace them with self-attention mechanisms.



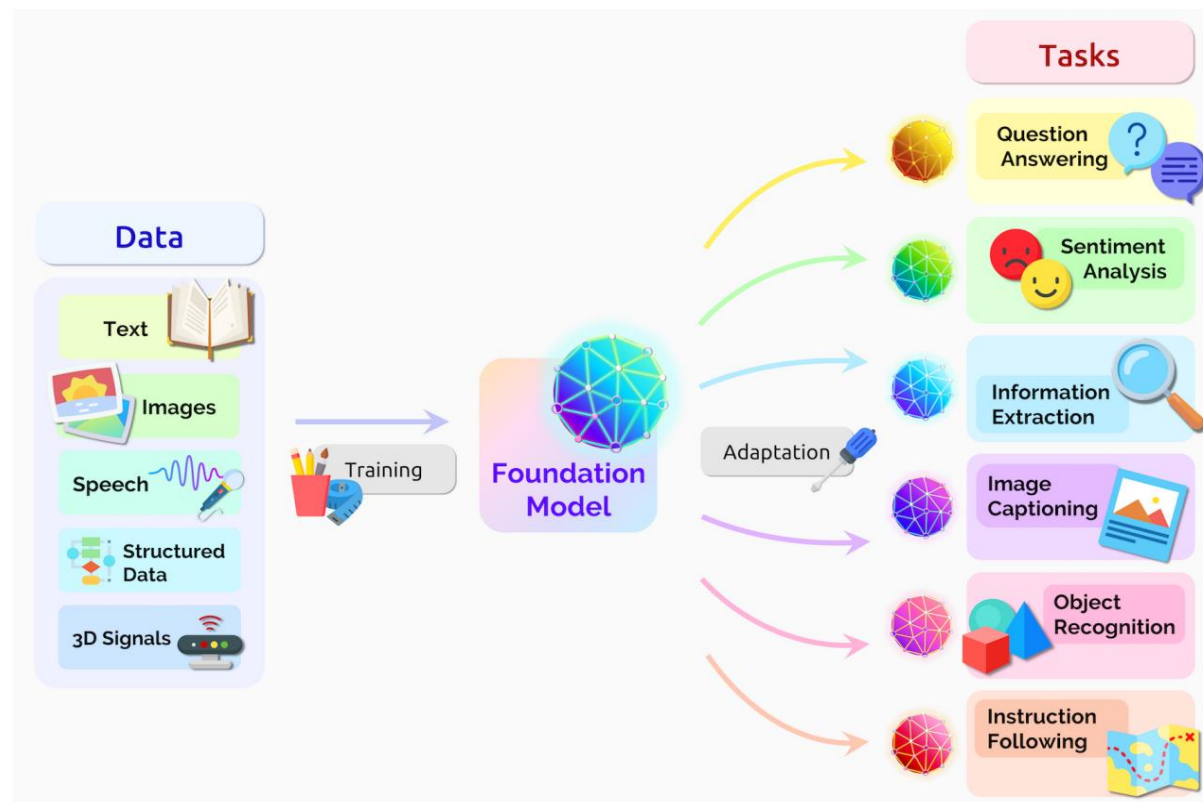
The increasing amount of Data Dilemma

- ✓ The increasing amount of data requires training AI models at a very large scale, therefore **Transformers** launch new era of **parameter-driven AI**. Learning to understand how hundreds of millions of words fit together in sentences requires a tremendous amount of parameters.
- ✓ **Transformers encoders** and **decoders** contain **attention heads** that train separately, parallelizing cutting-edge hardware. Attention heads can run on separate **GPUs** opening the door to billion-parameter models and soon-to-come trillion-parameter models. **OpenAI** trained a 175 billion parameter **GPT-3** Transformer model on a supercomputer with **10,000 GPUs** and **285,000 CPU** cores.



What are Foundation Models?

- ✓ **Foundation Models** are fully trained transformer models that can perform hundreds of tasks even without fine-tuning. Foundation models at this scale offer the tools we need in this massive information era.
- ✓ In the field of **Sentiment Analysis** for example, how many humans it would take to read the content of billions of tweets or posts on social networks in a few minutes?
- ✓ How many humans would be needed to translate the millions of **pages published** each day on the web?



- ✓ How many humans it would take to write the transcripts of all of the vast amount of hours of **streaming published** per day on the web?
- ✓ How many humans are required to replace **AI image captioning** for the billions of images that continuously appear online?

AI 2.0 and Industry 4.0

- ✓ **Transformers** are **industrialized, homogenized** deep learning models designed for **parallel** computing on **supercomputers**.
- ✓ Thanks to **homogenization**, one transformer model addresses a wide range of tasks with no fine-tuning. Transformers can perform self-supervised learning on billions of records of raw unlabeled data with billions of parameters.
- ✓ These particular architectures are called foundation models. **Foundation model** transformers represent the edge of the **Fourth Industrial Revolution** that began in **2015** with machine-to-machine automation that will connect everything to everything.
- ✓ **Artificial intelligence** in general and specifically **Natural Language Processing (NLP)** for **Industry 4.0 (I4.0)** has gone far beyond the software practices of the past.



- ✓ An **Industry 4.0** Project Manager may go to OpenAI's cloud platform, sign up, obtain an **API key**, and get to work in a few minutes. On the other side, a user can enter a text, specify the NLP task, and obtain a response sent by a GPT-3 transformer.
- ✓ Finally, a user can go to **GPT-3 Codex** and develop programs without any coding knowledge.

AI 2.0 and Industry 4.0

- ✓ A GPT-3 model might not fit a specific task. For example, a project manager, consultant, or developer might want to use another system provided by Google AI, Amazon Web Services (AWS), the Allen Institute for AI, or Hugging Face.
- ✓ Should a Project Manager choose to work locally? Or should the implementation be done directly on Google Cloud, Microsoft Azure, or AWS? Yes
- ✓ Should a development team select Hugging Face, Google Trax, OpenAI, or AllenNLP? Yes
- ✓ Should an **Artificial Intelligence** specialist or a **Data Scientist** use an API with practically no AI development? Yes



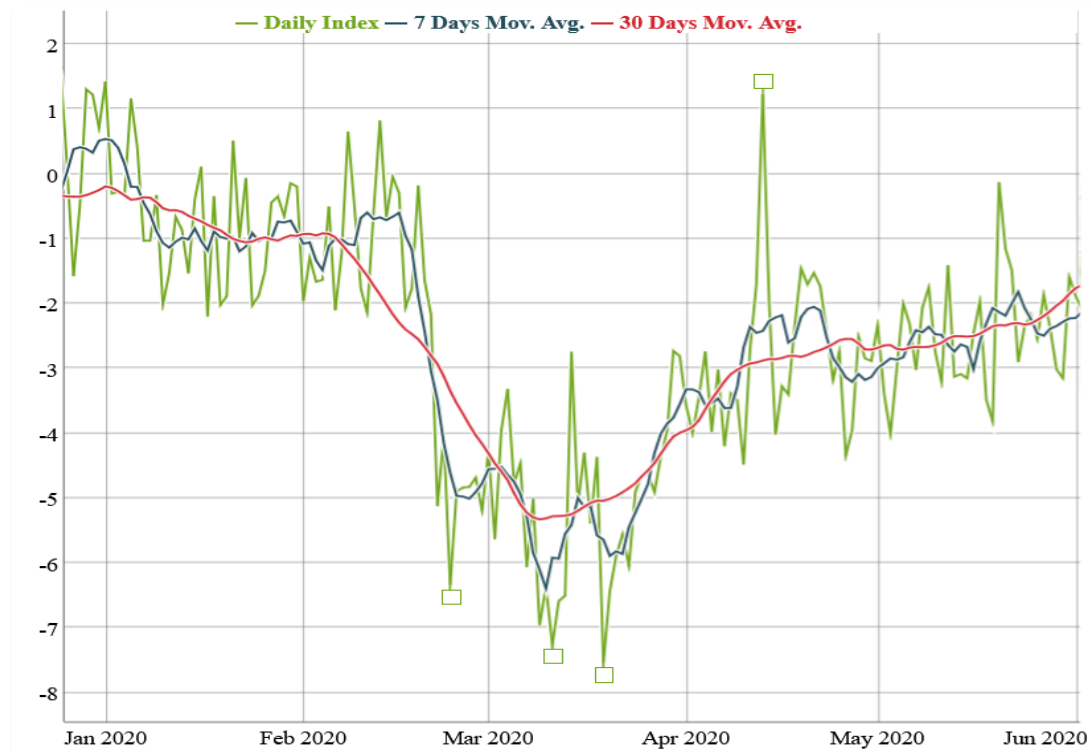
HUGGING FACE



Use Case 1: AI for Sentiment Analysis

- The Social Mood on Economy Index (SMEI) is developed by ISTAT
- Measures the Italian sentiment on economy
- Extracts the sentiment from Twitter data
- The sentiment analysis is conducted using a lexicon-based, unsupervised, approach

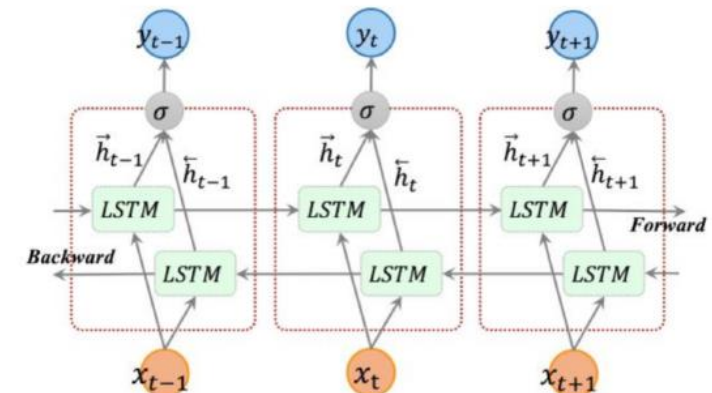
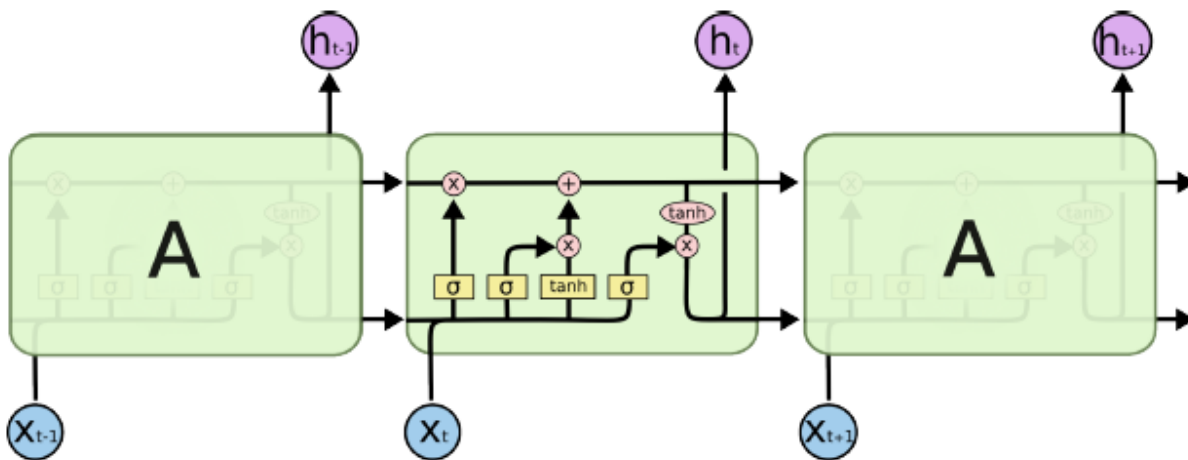
ISTAT's SMEI January-May 2020: the Covid outbreak in Italy



Use Case 1: AI for Sentiment Analysis

Bidirectional LSTM

- Our classification model is a Bidirectional Long short-term memory (BiLSTM). LSTMs are a type of RNN able to process long sequences of data whereas BiLSTM are able to understand the left context and right context of sequences.
- A LSTM memory cell is composed of 4 units exactly: an input gate, an output gate, a forget gate and a self-recurrent neuron.



Use Case 1: AI for Sentiment Analysis

FastText

- Our **Word Embeddings Model** (*Catanese, E., Bruno, M., Stefanelli, L., & Pugliese, F., 2023, September - Winner of the Best Paper Award - 1st Place - CARMA 2023*) is **FastText** which is an evolution of **Word2Vec**.
- **FastText** enables to quickly train models on large corpora
- **FastText** make use of hierarchical soft-max based which is able to noticeably drop computational complexity and can be trained on more than one billion words in less than ten minutes using a standard multicore **CPU**.



Use Case 1: AI for Sentiment Analysis

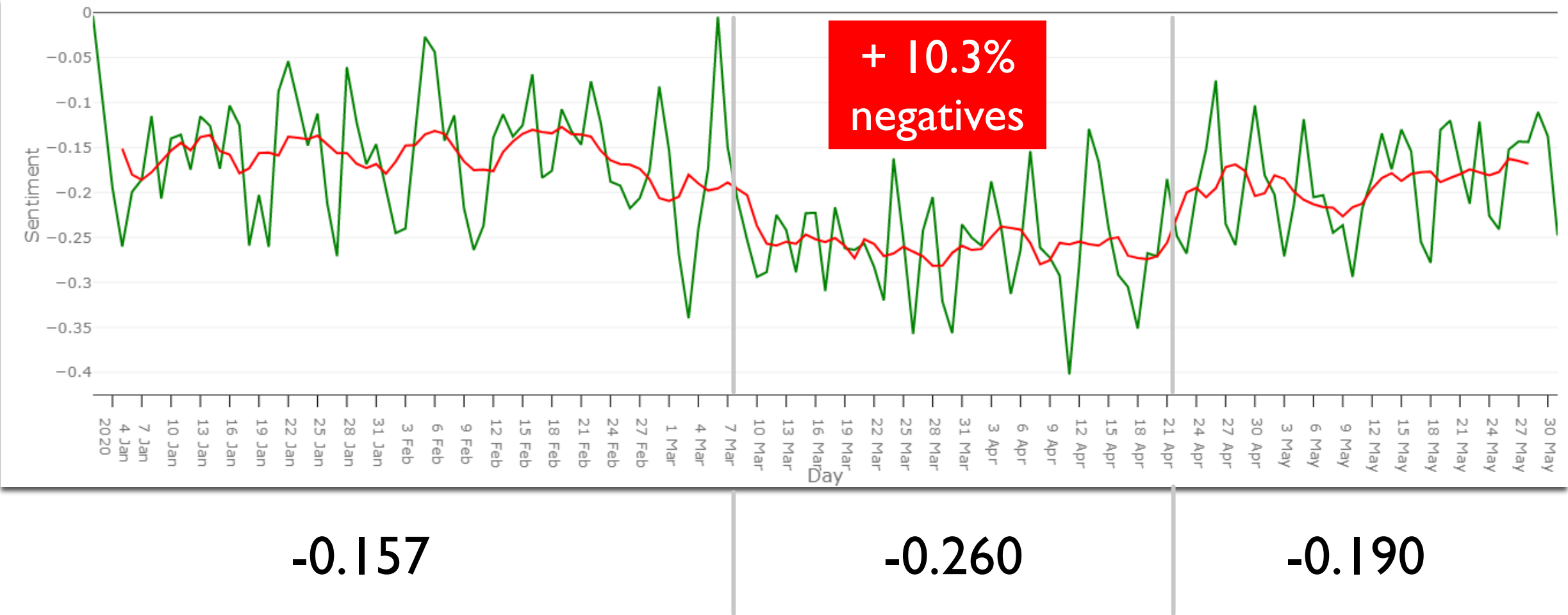
Why a new Index?

- ISTAT's Index (SMEI) needed a revision to deal with new events (Covid pandemic)
 - The proposed model may have a higher adaptability to new information
 - The proposed model is a neural network that has been trained on a set of labelled Tweets that do not contain references to the pandemic
- Computed daily using the sentiment predicted by the model on a set of tweets
 - The sentiment is predicted using a neural network
 - The set of tweets is the same used by ISTAT for the SMEI
 - Computed for the first half of 2020

$$DL - SMEI = \frac{N_P - N_N}{N_P + N_N}$$

Use Case 1: AI for Sentiment Analysis

The proposed Index (DL-SMEI)



Use Case 1: AI for Sentiment Analysis

1st January – 7th March



8th March – 21st April



22nd April – 31st May



Use Case 1: AI for Sentiment Analysis

Focus on second period (8th March – 21st April)

○ Positive words

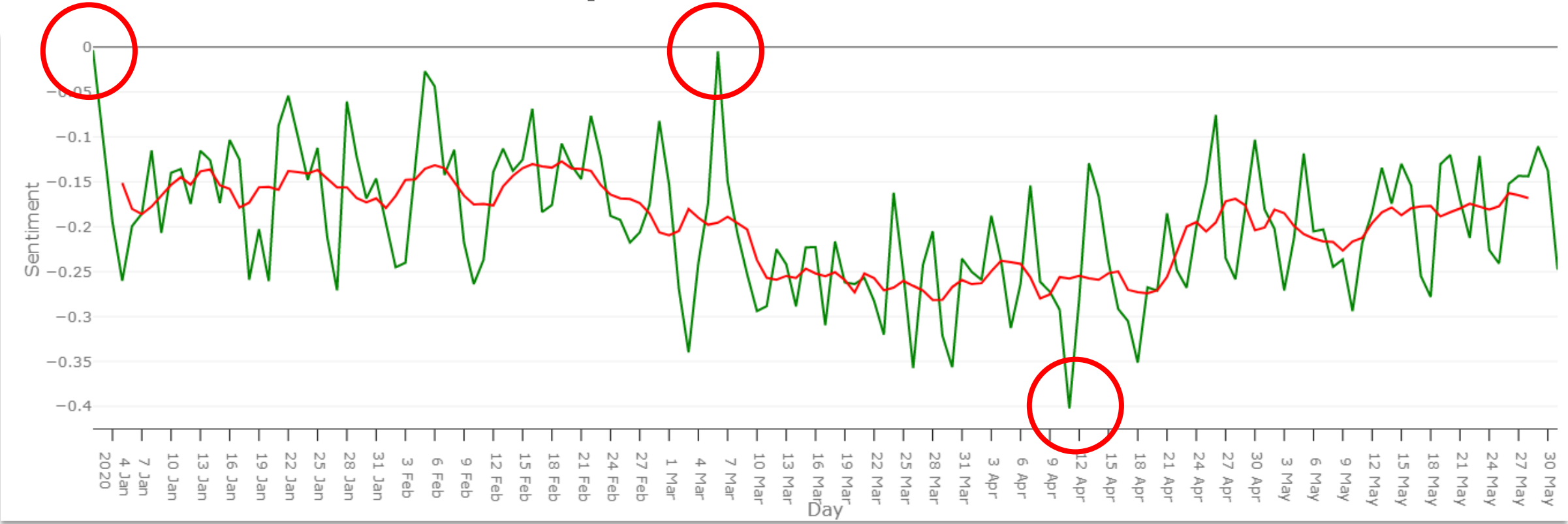


Negative words



Use Case 1: AI for Sentiment Analysis

Maximum and Minimum points



Use Case 1: AI for Sentiment Analysis

Second Maximum on the 6th March

○ Positive words



Negative words



Use Case 2: AI for Land Cover

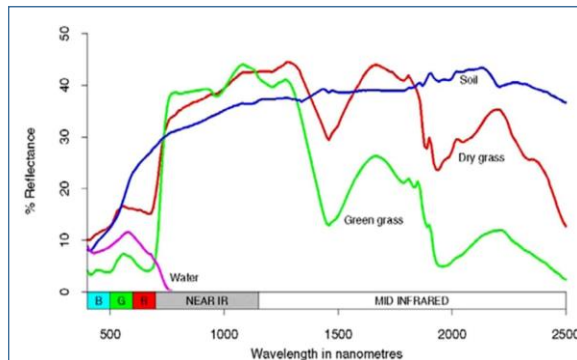
GOALS

Land Cover (LC) statistics (*Bernasconi, E., De Fausti, F., Pugliese, F., Scannapieco, M., & Zardetto, D., 2022*) and maps are a very important statistical product. As they require a big effort to be created, the idea is to build an automatic system that processes satellite

- Automatic Land Cover Estimates
- Automatic Land Cover Maps

ML Approaches to LC from Images

Standard approach: Spectral Signature



- Different LC classes have different reflectance spectra
 - ✓ Variation of reflectance with EM frequency can be used to predict LC class
 - ✓ Trained ML algo predicts the LC class of image pixels independently
 - ✓ Decision on each pixel does not depend on neighboring pixels

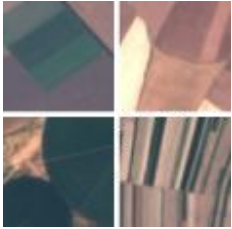
New approach: Computer Vision (Deep Learning)



- Different LC classes have different visual/spatial patterns
 - ✓ Variation of visual/spatial patterns can be used to predict LC class
 - ✓ Trained ML algo (CNN/U-net) predicts LC class of image pixels based on information from neighboring pixels
 - ✓ Decision on each pixel depends on the whole sub-image (tile) the pixel belongs to

Use Case 2: AI for Land Cover

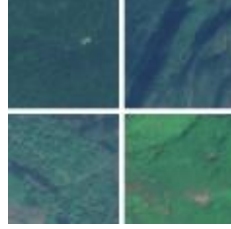
ANNUAL CROP



RIVER



FOREST



RESIDENTIAL



INDUSTRIAL



HIGHWAY



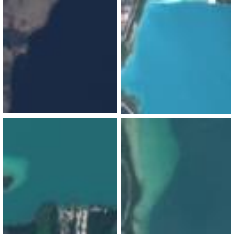
PASTURE



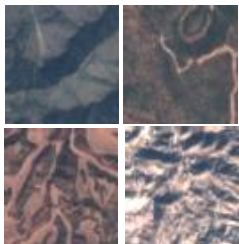
PERMANENT CROP



SEA LAKE



HERBACEOUS VEGETATION



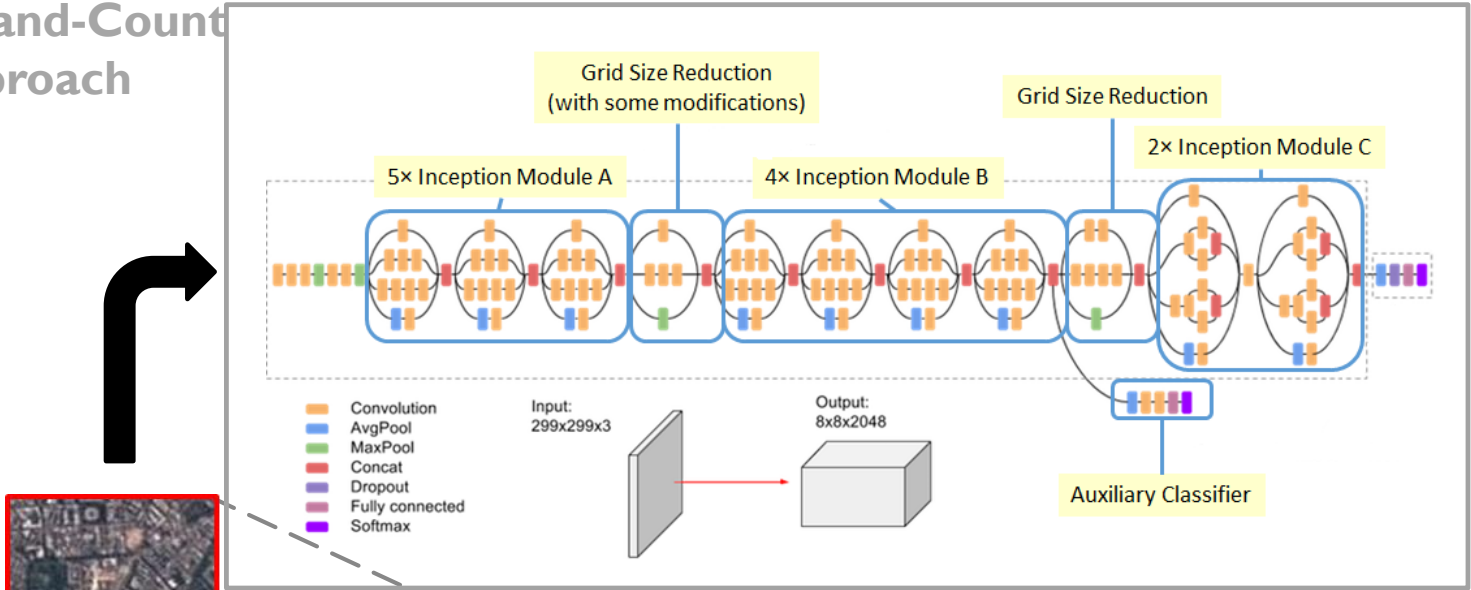
EuroSAT dataset

(<https://github.com/phelber/eurosat>):

- Based on Sentinel-2 satellite images
- 27000 geo-referenced and labeled image patches (each one of 64x64 pixels)
- 10 different Land Use and Land Cover classes, with 2000-3000 images per class
- RGB (8-bit) and Multi-Spectral (13 spectral bands, 16-bit) versions available

Use Case 2: AI for Land Cover

Classify-and-Count
Approach

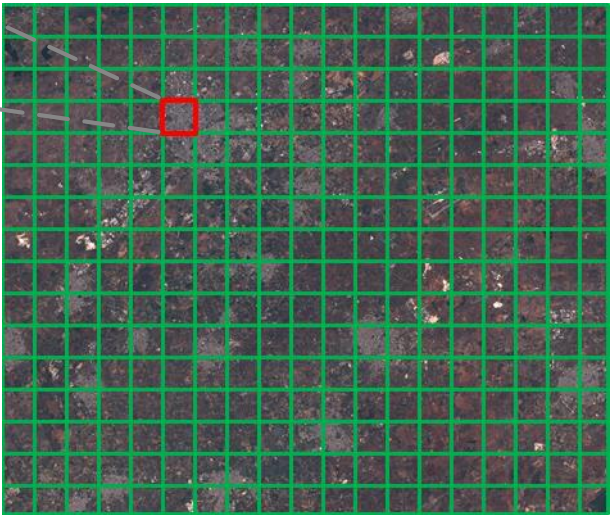


CNN: Inception-V3 Archite

RESIDENTIAL

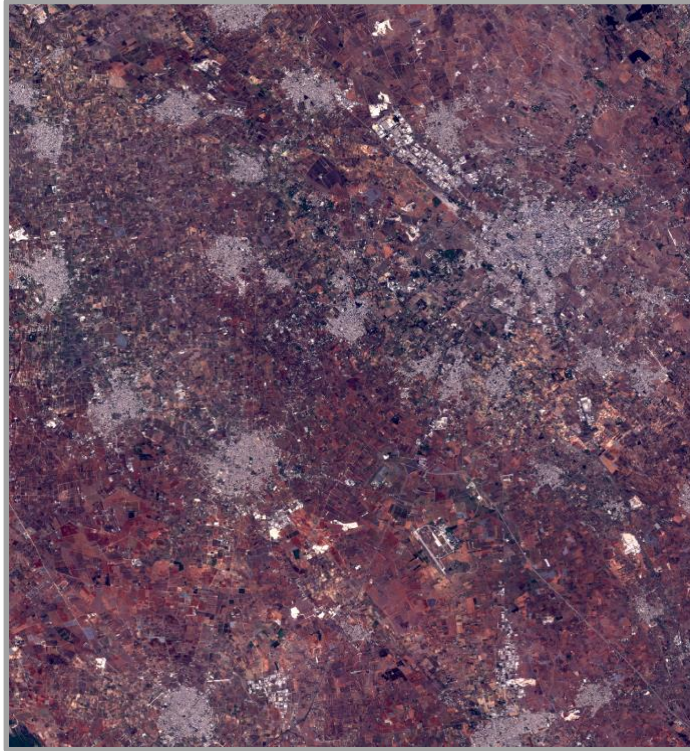


Input Satellite
Image



LAND COVER CLASS	AREA SHARE
...	...
RESIDENTIAL	$\frac{45}{16 * 19} \cong 15\%$
...	...

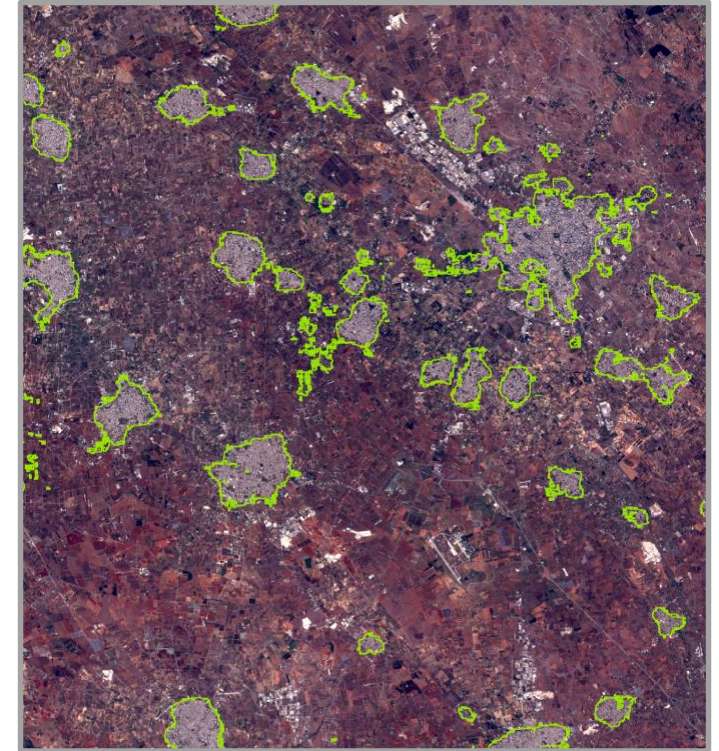
Use Case 2: AI for Land Cover



[A]
The ‘**Lecce image**’
(751 km²)



[B]
Automated LC map derived
from the ‘Lecce image’



[C]
Edge line of the ‘Residential’
class derived from [B] overlaid
on [A]

Use Case 2: AI for Land Cover



[D]

A detailed view of the course of the Arno River (cropped from the **'Pisa image'**, 443 km²) overlaid with a semitransparent version of the corresponding automated LC map

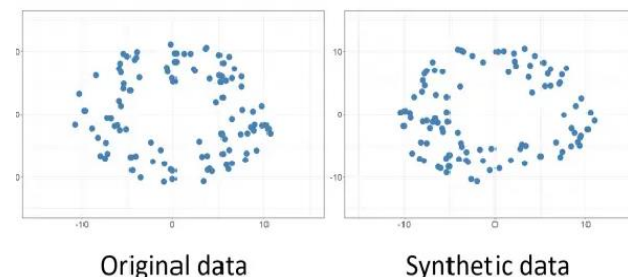


[E]

A highway fragment from the **'Lecce image'** overlaid with the edge line of the 'Highway' class

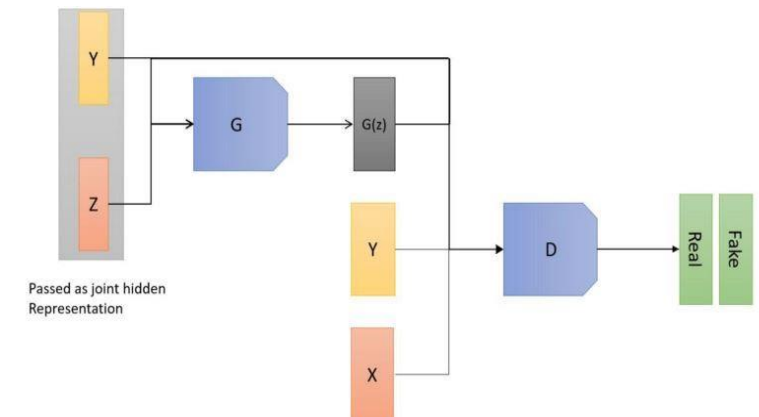
Use Case 3: AI for the generation of Synthetic Data

- ✓ With the **digitalization of information** and the increasing accessibility of **administrative data**, the amount of data to be handled has grown substantially in recent years. This raises significant concerns about **data protection** and **privacy** since the disclosure of sensitive information can pose serious risks to individuals, institutions, and public administrations.
- ✓ **Synthetic data** (*Pugliese, F., Pappagallo, A., & De Cubellis, M., 2024, June*) are artificially created datasets intended to **replicate** the statistical characteristics and structure of real data, while preventing the exposure of **sensitive** or **personally identifiable** information.



Use Case 3: AI for the generation of Synthetic Data

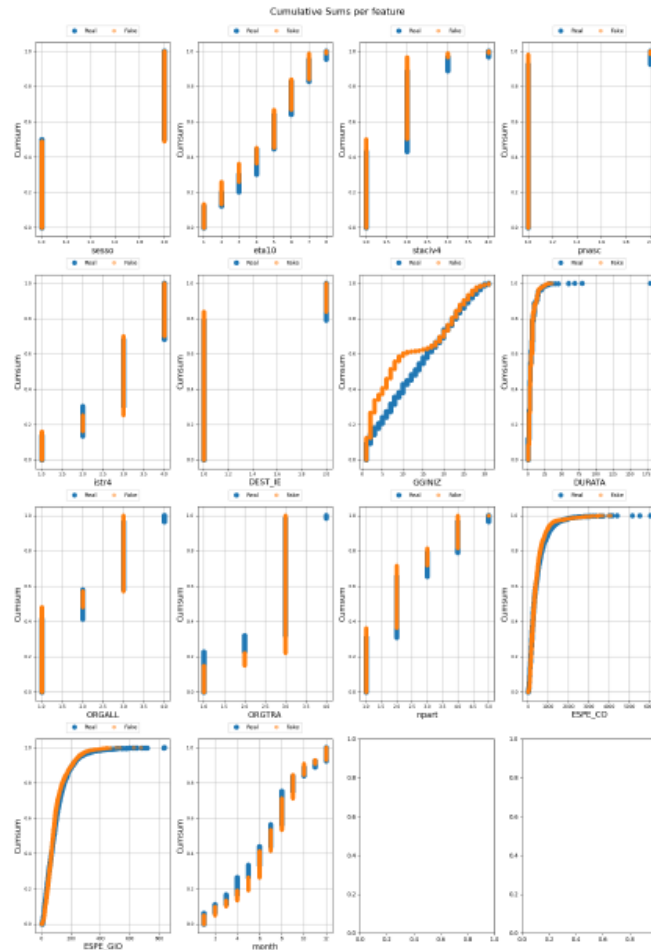
- ✓ **Keep in mind!** Synthetic data do not necessarily reproduce the atomic data (categorical, numerical, etc) from the original source. **Synthesis** capability might rely rather on the **relationships** between different kind of **entities**, such as people and objects, school and neighborhoods, or users and cellular antennas.
- ✓ In this study, the **analyzed data** relate to the frequency and attributes of trips and vacations undertaken by residents of Italy. They originate from **the Istat Trips and Holidays survey**, as a module of the **Household Budget Survey (HBS)** which collects information on the tourism flows of residents. This information includes journeys made for leisure or work, both domestically and internationally.
- ✓ In the WPI3's Istat PoC of AI/ML Essnet project, we compared different AI methods: **CT-GAN, VAEs, DP-CTGAN, SMOTENC, Random Forest, XGBoost**



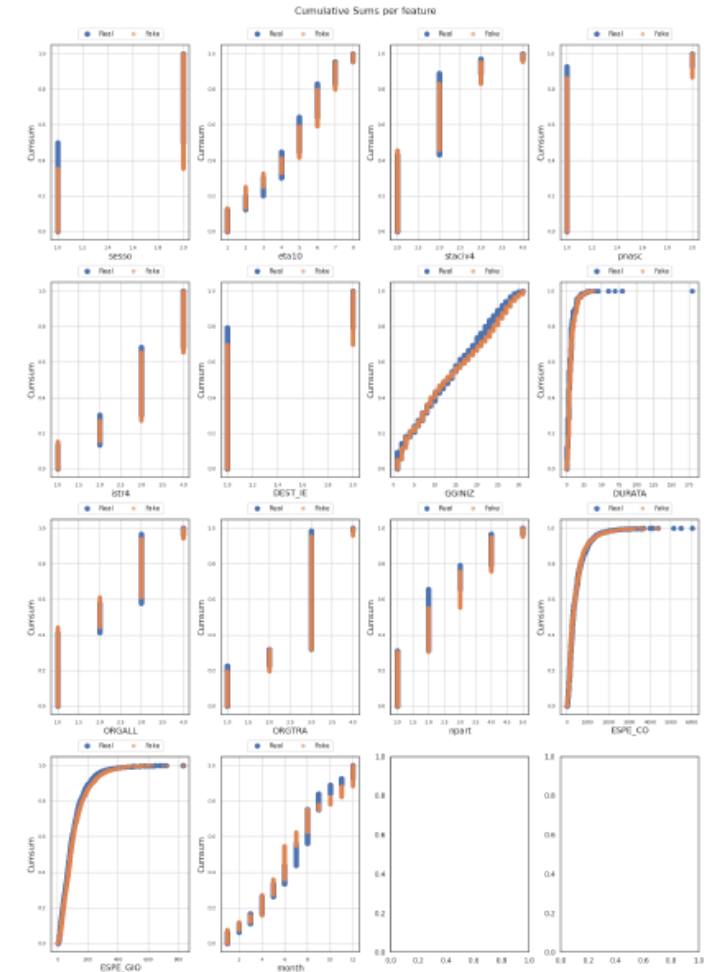
Use Case 3: AI for the generation of Synthetic Data

- ✓ Note: In the **original distribution itself**, the values are *not* sums of previous values. Only in the **cumulative sum version** do the values accumulate.
- ✓ It is **evident** from the **cumulative** distributions that the less effective methods, such as **Random Forest** or **XGBoost**, are **less** efficient on categorical variables, where they often **fail** to reproduce the categories, i.e. the domain values of these variables.

Variational Autoencoder (VAE)



CT-GAN



Use Case 3: AI for the generation of Synthetic Data

- ✓ As we can observe, apart from **SMOTENC** which tends to reproduce the data exactly, the best methods seem to be the deep learning–based ones such as CTGAN and VAE, as they achieve high accuracy but not identical to the original dataset, as expected. In fact, during the synthetic data generation process something is always **lost** in terms of the **properties** of the data being reproduced.

Model	Accuracy	F1-Score	Recall
Original Data	0.964516848	0.964516848	0.964516848
RF	0.660685592	0.660685592	0.660685592
XGB	4.46349e-05	4.46349e-05	4.46349e-05
SMOTENC	0.992813783	0.992813783	0.992813783
DPCTGAN	4.46349e-05	4.46349e-05	4.46349e-05
VAE	0.892385288	0.892385288	0.892385288
CTGAN	0.896134619	0.896134619	0.896134619

Use Case 4: AI for the automatic classification of Causes of Death

Detection Model

For each individual death occurring within the national territory, the certifying physician records the information on the causes of death in this part of the paper certificate

ATTENZIONE

Leggere le ISTRUZIONI sul RETRO

Stampare in STAMPATELLO

ISTITUTO NAZIONALE DI STATISTICA

SCHEDA DI MORTE OLTRE IL 1° ANNO DI VITA

ISTAT MODELLO 4 - E66 2011

COPIA PER L'ISTAT

PARTE A - A CURA DEL MEDICO

MASCHIO

FEMMINA

Cognome e nome della persona defunta

1. Luogo del decesso

Indirizzo

Indirizzo di un pubblico carcere o assimilato

Indirizzo

Struttura ospedaliera o socio-sanitaria

Altro (specificare)

Città

Provincia

Codice ABL

2. Data e ora di avvenimento

Giorno

Mese

Anno

Orario

3. Ricovero diagnostico

C'era ricovero?

XX

Causa Brodie

Indirizzo di nascita

XX

4. Esito clinico

MASCHIO

FEMMINA

5. Cause

1. Data di morte

2. Data di nascita

3. Località di nascita

4. Esito clinico

5. Data di morte

6. Data di nascita

7. Località di nascita

8. Esito clinico

9. Data di morte

10. Data di nascita

11. Località di nascita

12. Esito clinico

13. Data di morte

14. Data di nascita

15. Località di nascita

16. Esito clinico

17. Data di morte

18. Data di nascita

19. Località di nascita

20. Esito clinico

21. Data di morte

22. Data di nascita

23. Località di nascita

24. Esito clinico

25. Data di morte

26. Data di nascita

27. Località di nascita

28. Esito clinico

29. Data di morte

30. Data di nascita

31. Località di nascita

32. Esito clinico

33. Data di morte

34. Data di nascita

35. Località di nascita

36. Esito clinico

37. Data di morte

38. Data di nascita

39. Località di nascita

40. Esito clinico

41. Data di morte

42. Data di nascita

43. Località di nascita

44. Esito clinico

45. Data di morte

46. Data di nascita

47. Località di nascita

48. Esito clinico

49. Data di morte

50. Data di nascita

51. Località di nascita

52. Esito clinico

53. Data di morte

54. Data di nascita

55. Località di nascita

56. Esito clinico

57. Data di morte

58. Data di nascita

59. Località di nascita

60. Esito clinico

61. Data di morte

62. Data di nascita

63. Località di nascita

64. Esito clinico

65. Data di morte

66. Data di nascita

67. Località di nascita

68. Esito clinico

69. Data di morte

70. Data di nascita

71. Località di nascita

72. Esito clinico

73. Data di morte

74. Data di nascita

75. Località di nascita

76. Esito clinico

77. Data di morte

78. Data di nascita

79. Località di nascita

80. Esito clinico

81. Data di morte

82. Data di nascita

83. Località di nascita

84. Esito clinico

85. Data di morte

86. Data di nascita

87. Località di nascita

88. Esito clinico

89. Data di morte

90. Data di nascita

91. Località di nascita

92. Esito clinico

93. Data di morte

94. Data di nascita

95. Località di nascita

96. Esito clinico

97. Data di morte

98. Data di nascita

99. Località di nascita

100. Esito clinico

PARTE B - A CURA DELL'UFFICIALE DI STATO CIVILE (controllare ed eventualmente correggere cognome e nome)

MASCHIO

FEMMINA

Codice FISCALE della defunta

1. Data di morte

2. Data di nascita

3. Località di nascita

4. Esito clinico

5. Data di morte

6. Data di nascita

7. Località di nascita

8. Esito clinico

9. Data di morte

10. Data di nascita

11. Località di nascita

12. Esito clinico

13. Data di morte

14. Data di nascita

15. Località di nascita

16. Esito clinico

17. Data di morte

18. Data di nascita

19. Località di nascita

20. Esito clinico

21. Data di morte

22. Data di nascita

23. Località di nascita

24. Esito clinico

25. Data di morte

26. Data di nascita

27. Località di nascita

28. Esito clinico

29. Data di morte

30. Data di nascita

31. Località di nascita

32. Esito clinico

33. Data di morte

34. Data di nascita

35. Località di nascita

36. Esito clinico

37. Data di morte

38. Data di nascita

39. Località di nascita

40. Esito clinico

41. Data di morte

42. Data di nascita

43. Località di nascita

44. Esito clinico

45. Data di morte

46. Data di nascita

47. Località di nascita

48. Esito clinico

49. Data di morte

50. Data di nascita

51. Località di nascita

52. Esito clinico

53. Data di morte

54. Data di nascita

55. Località di nascita

56. Esito clinico

57. Data di morte

58. Data di nascita

59. Località di nascita

60. Esito clinico

61. Data di morte

62. Data di nascita

63. Località di nascita

64. Esito clinico

65. Data di morte

66. Data di nascita

67. Località di nascita

68. Esito clinico

69. Data di morte

70. Data di nascita

71. Località di nascita

72. Esito clinico

73. Data di morte

74. Data di nascita

75. Località di nascita

76. Esito clinico

77. Data di morte

78. Data di nascita

79. Località di nascita

80. Esito clinico

81. Data di morte

82. Data di nascita

83. Località di nascita

84. Esito clinico

85. Data di morte

86. Data di nascita

87. Località di nascita

88. Esito clinico

89. Data di morte

90. Data di nascita

91. Località di nascita

92. Esito clinico

93. Data di morte

94. Data di nascita

95. Località di nascita

96. Esito clinico

97. Data di morte

98. Data di nascita

99. Località di nascita

100. Esito clinico

PARTE C - A CURA DELL'UFFICIALE DI STATO CIVILE (controllare ed eventualmente correggere cognome e nome)

MASCHIO

FEMMINA

Codice FISCALE della defunta

1. Data di morte

2. Data di nascita

3. Località di nascita

4. Esito clinico

5. Data di morte

6. Data di nascita

7. Località di nascita

8. Esito clinico

9. Data di morte

10. Data di nascita

11. Località di nascita

12. Esito clinico

13. Data di morte

14. Data di nascita

15. Località di nascita

16. Esito clinico

17. Data di morte

18. Data di nascita

19. Località di nascita

20. Esito clinico

21. Data di morte

22. Data di nascita

23. Località di nascita

24. Esito clinico

25. Data di morte

26. Data di nascita

27. Località di nascita

28. Esito clinico

29. Data di morte

30. Data di nascita

31. Località di nascita

32. Esito clinico

33. Data di morte

34. Data di nascita

35. Località di nascita

36. Esito clinico

37. Data di morte

38. Data di nascita

39. Località di nascita

40. Esito clinico

41. Data di morte

42. Data di nascita

43. Località di nascita

44. Esito clinico

45. Data di morte

46. Data di nascita

47. Località di nascita

48. Esito clinico

49. Data di morte

50. Data di nascita

51. Località di nascita

52. Esito clinico

53. Data di morte

54. Data di nascita

55. Località di nascita

56. Esito clinico

57. Data di morte

58. Data di nascita

59. Località di nascita

60. Esito clinico

61. Data di morte

62. Data di nascita

63. Località di nascita

64. Esito clinico

65. Data di morte

6

4. Parte I	CAUSA DI MORTE: sequenza di condizioni morbose o traumi o avvelenamenti che ha condotto a morte - In presenza di più sequenze scegliere la più rilevante	Tempo intercorso tra l'insorgenza della causa indicata e la morte
Causa iniziale. Scegliere la SOLA patologia o trauma che ha dato inizio alla sequenza.	1 ↓ ↓ che ha causato	____ anni ____ mesi ____ giorni
EVENTUALI condizioni o complicazioni che fanno parte della sequenza che ha portato a morte.	2 ↓ ↓ che ha causato	____ anni ____ mesi ____ giorni
	3 ↓ ↓ che ha causato	____ anni ____ mesi ____ giorni
	4 	____ anni ____ mesi ____ giorni
4. Parte II	Altri stati morbose rilevanti - Indicare altre malattie o condizioni morbose o traumi esclusi dalla sequenza riportata nel quesito 4. Parte I, ma che hanno contribuito al decesso.	
		____ anni ____ mesi ____ giorni
		____ anni ____ mesi ____ giorni
		____ anni ____ mesi ____ giorni
N.B. I QUESITI 5-9 DEVONO ESSERE COMPILATI SOLO IN CASO DI MORTE DA TRAUMATISMO/AVVELENAMENTO (anche con informazioni riferite)		
5. Circostranza che ha dato origine al trauma/avvelenamento indicato in 4. Parte I (es. caduta da scala a pioli, impiccamento, fucilata, ecc.)		
6. Modalità del trauma/avvelenamento	8. Data dell'accidente, infortunio, suicidio, omicidio	
Accidentale (include incidenti da trasporto).....1 ☐	Suicidio.....2 ☐ Omicidio.....3 ☐	
6.1 Infortunio sul lavoro 1 ☒ 2 ☐	____ Orario ____ giorno ____ mese ____ anno	

Istat Form D4-D4bis 'Death Certificate'

Use Case 4: AI for the automatic classification of Causes of Death

The project aims to:

- identify the most suitable AI methods for coding causes of death;
- create an algorithm for coding causes of death by applying these methods;
- evaluate the performance of automatic coding and the quality of the data by applying this algorithm.

The main expected results are:

- a reduction in coding variability among coders and in errors, resulting in improved quality;
- in view of the imminent replacement of the paper form with an electronic model, AI methods will be useful to maintain high performance in recognizing medical text, even in the absence of the preprocessing currently applied during the registration of paper forms.
- an increase in the number of death certificates coded completely automatically, without human intervention;

Use Case 4: AI for the automatic classification of Causes of Death

The information on the causes is recorded by physicians on a paper death certificate, digitized, and coded according to the **tenth revision of the International Classification of Diseases (ICD-10)** of the **World Health Organization (WHO)**.

The application of the ICD is a complex process that involves:

- first, assigning a code to all the causes certified by physicians on the forms (multiple causes).
- then, selecting only one code for each death: the **underlying cause**.

Both phases are carried out according to the rules described in the ICD, and both the multiple causes and the underlying cause are used for mortality statistics.

Use Case 4: AI for the automatic classification of Causes of Death

Training set 22_23: about 4 million texts.

53,876 classes after class cleaning and drops with cutoff.

Too much data for training: we sample progressively (starting from 300,000 texts).

DS_Gold_standard (external): 951 texts assigned a code whose correctness has been verified.

TEST_08_09 (test_set_blind): 11,909 texts to be labeled.

Valid ICD codes

Flag (A, C, CON, P, etc). Example:

Separators (;, /, etc). Example: *CACHESSIA TERMINALE CON IRA IN GRANDE ANZIANO* coded as **NI79 / R54**.

(Note: “*Cachessia terminale con IRA in grande anziano*” = “*Terminal cachexia with acute renal failure in a very elderly patient.*”)

Use Case 4: AI for the automatic classification of Causes of Death

TF-IDF on the dataset to obtain the normalized term-document matrix; **Random Forest** is used for classification (which works on SL).

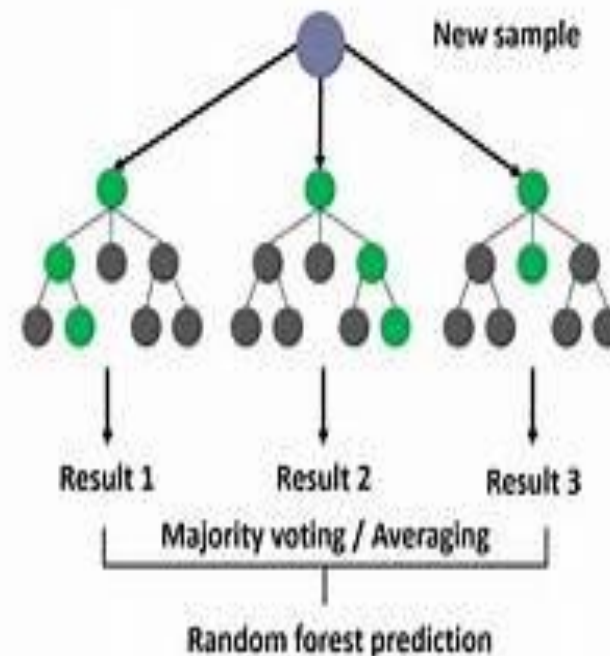
We take a sample of **300,000 texts** for training.

Initially **54,767 classes**, later reduced to **27,364**, after non-semantic splitting (in cases where the number of classes equals the number of texts).

We use a **cutoff of 2**, meaning we consider only classes with at least two instances.

There are about **17,666 classes** that appear only once, and these are removed.

In the end, we are left with **9,698 classes**.



Use Case 4: AI for the automatic classification of Causes of Death

On training set of 300,000:

Accuracy: 0.9761%

F1-score (macro-weighted): 0.9745%

On test set (20%):

Accuracy: 0.9467%

F1-score (macro-weighted): 0.9422%

Accuracy on test set of about 65,161 texts:

- #codes = #texts (size = 63,636, 97.66% of the test dataset): Accuracy: 0.9528% – F1-score (macro-weighted): 0.9505%
- #codes > #texts (size = 1,525, 2.34% of the test dataset): Accuracy: 0.69% – F1-score (macro-weighted): 0.71% → 1,060 correct, 465 incorrect (outputs exported to Excel and CSV for sharing)
- #texts > #codes (size = 0)

(Note: The “%” signs seem to be used instead of decimals — e.g., accuracy 0.9761 likely means 97.61%.)

Use Case 4: AI for the automatic classification of Causes of Death

Accuracy on test set of about 65,161 texts:

#codes > #texts (size = 1,525, 2.34% of the test dataset): Accuracy: 0.69% – F1-score (macro-weighted): 0.71% → 1,060 correct, 465 incorrect.

The correct cases are mostly those in which one text corresponds to two codes.

We check what percentage of the texts corresponding to the correct cases are present in the training set: **915 out of 1,050 (86%)**.

We check what percentage of the texts corresponding to the incorrect cases are present in the training set: **76 out of 475 (16%)**.

Theory: Random Forest performs well when it finds the same words combined in the same way in the training set.

Use Case 4: AI for the automatic classification of Causes of Death

Accuracy on test set (sample of 950 “gold_standard” texts):

For this dataset, the initial number of classes is **876**. After non-semantic splitting (in cases where the number of classes equals the number of texts — i.e., for 197 cases), we obtain **847 classes**, and the number of texts to be classified becomes **1,154**.

The classes are reduced to **716** after removing the time period.

Accuracy on “gold_standard” test set: overall **0.41% F1-score** → problem: only **24% of the gold standard texts** are present in the training set, while **62% of the classes** are present in the training set.

- #codes = #texts (size = 948, 82.15% of the gold standard dataset): Accuracy: 0.55% – F1-score (macro-weighted): 0.52%

- #codes > #texts (size = 206, 17.85% of the gold standard dataset): Accuracy: 0.033% – F1-score (macro-weighted): 0.033% → 7 correct, 199 incorrect (outputs exported to Excel and CSV for sharing)

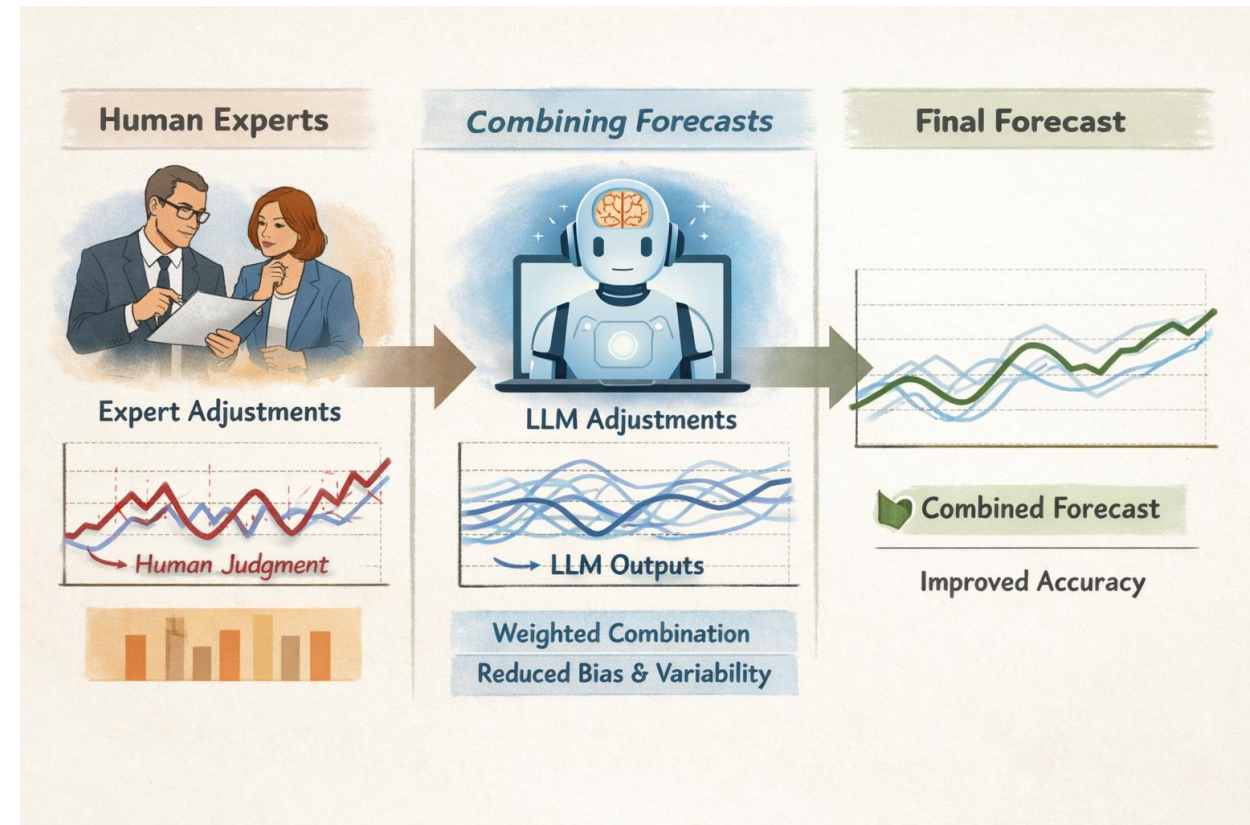
- #texts > #codes (size = 0)

OBSERVATION: for the label encoding of the classes, we include both the classes present in the training set and those in the gold standard, to prevent cases where classes in the gold standard are missing from the training data.

In the previous case, this issue did not arise because we had obtained it using a **stratified train-test split**, which ensures that each class has a sample in both the training and test sets.

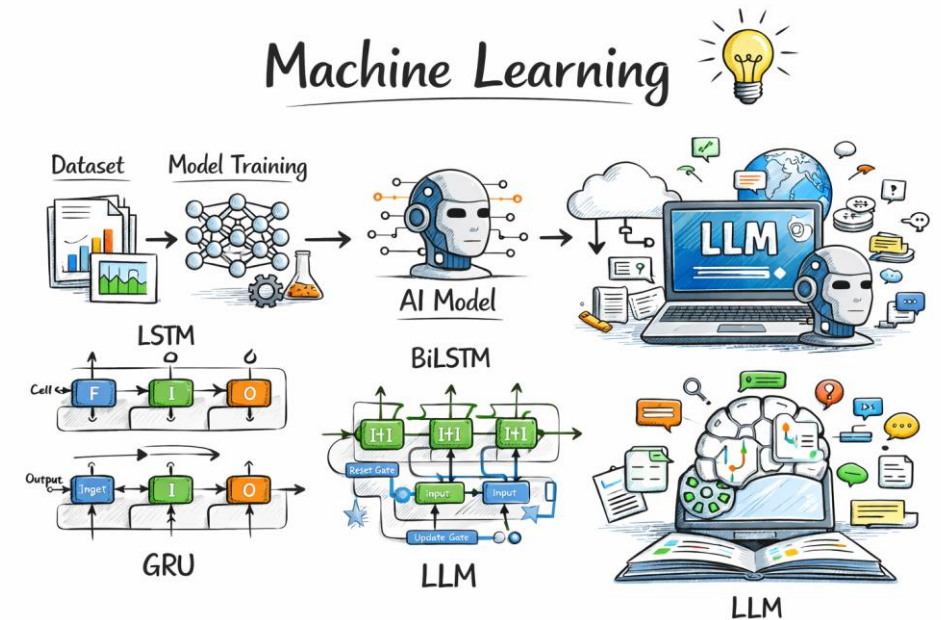
Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ **Statistical models** give consistent forecasts but often **miss context**, structural changes, and expert knowledge.
- ✓ To compensate, practitioners **adjust forecasts** using **human judgment**, though these adjustments can introduce biases and inconsistencies.
- ✓ **Research** shows that combining multiple adjusted **forecasts** — especially with simple methods like equal weighting — **improves accuracy** by reducing individual biases and extreme errors.



Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ Building on this idea, in this study we propose replacing human judgment with a **Large Language Model (LLM)**. Each LLM-generated adjustment is treated like an expert opinion, and multiple outputs are combined to improve robustness.
- ✓ This approach creates a **scalable, consistent** alternative to human experts and is applied to **Inflation** forecasting, where **judgment-based revisions** are common.
- ✓ After selecting the best-performing model (from **ARIMA** and Deep Learning methods like **LSTM, GRU, BiLSTM**, and **Transformers**), the LLM is queried multiple times to generate independent adjustments. These are combined to refine the baseline forecast.



Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ The **Harmonised Index of Consumer Prices (HICP)** measures inflation in the EU and euro area using a standardized basket of goods and services, ensuring comparability across countries under the methodology of Eurostat (*Eurostat - European Commission, Harmonised Index of Consumer Prices - HICP*).
- ✓ The index (base **2025 = 100**) reflects weighted household consumption and is regularly updated.
- ✓ It is **seasonally adjusted by the European Central Bank** to highlight underlying trends and is published monthly, serving as a key indicator for economic analysis and monetary policy decisions



Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ This Table presents the performance of both statistical and deep learning models across the full dataset, as well as the training and test subsets.
- ✓ Model evaluation was carried out using RMSE, MAE, and R2 Score, considering both rolling and non-rolling settings.

Dataset metrics			
Model	RMSE	MAE	R^2 score
Arima	10.94	1.696	0.887
ETS	11.866	2.158	0.872
SES	10.877	1.443	0.888
LSTM with rolling	45.312	28.543	-0.271
LSTM with no rolling	6.758	2.572	0.982
GRU with rolling	41.412	24.944	-0.119
GRU with no rolling	6.642	1.78	0.983
BiLSTM with rolling	42.937	26.053	-0.144
BiLSTM with no rolling	6.619	1.652	0.983
Vanilla Transformer with Rolling			
Vanilla Transformer with no Rolling	6.465	1.251	0.984

Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ Deep learning models showed superior performance when trained without rolling, adopting a conventional 80/20 split without shuffling. The prediction targets were obtained by shifting the time series forward, thereby framing the problem as a supervised learning task suitable for RNN-based models.
- ✓ Among all approaches, the Vanilla Transformer (VT) delivered the best overall results, while LSTM, GRU, and BiLSTM also achieved strong performance. Notably, VT exhibited the smallest gap between training and test suggesting lower overfitting and better generalization capability.

Training set metrics			
Model	RMSE	MAE	R^2 score
LSTM with no rolling	7.431	2.621	0.981
GRU with no rolling	7.224	1.661	0.982
BiLSTM with no rolling	7.182	1.482	0.982
Vanilla Transformer with no Rolling	7.215	1.322	0.982

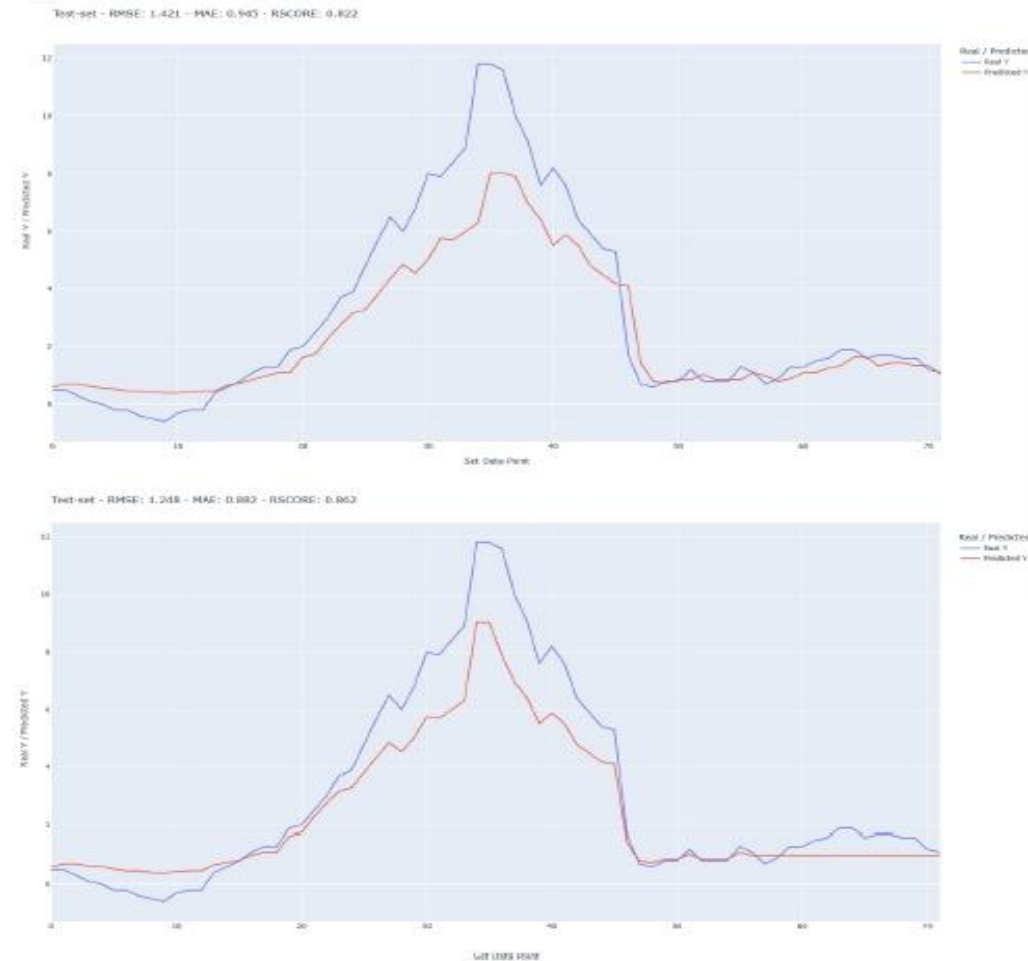
Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ Subsequently, a post-processing step was applied to VT predictions using LLaMA in a few-shot setting. The LLM, instructed as an expert in time series error correction, was provided with true training targets, VT training predictions and VT test predictions. This allowed the model to learn systematic prediction errors and generate refined corrections.
- ✓ This post-prediction refinement led to an improvement, highlighting the effectiveness of integrating deep learning models with LLM-based correction mechanisms.

Test set metrics				
Model	RMSE	MAE	R^2 score	Gap R^2 score Train-Test
LSTM with no rolling	2.828	2.379	0.294	0.687
GRU with no rolling	3.506	2.246	−0.101	1.083
BiLSTM with no rolling	3.634	2.324	−0.165	1.147
Vanilla Transformer with no Rolling	1.421	0.945	0.822	0.116
Vanilla Transformer with no Rolling adjusted by Llama	1.248	0.882	0.862	0.109(gain)

Use Case 5: AI for Time Series Forecasting by LLM-Based Adjustments

- ✓ The upper part of this figure illustrates the model prediction curves (in red) alongside the actual Test Set values. The top panel shows the predictions produced by the Vanilla Transformer, with the corresponding evaluation metrics reported above.
- ✓ In the lower panel, the predictions after LLM-based adjustment are presented, revealing a smoother curve that more closely aligns with the true values. This improvement is consistent with the reported metrics.



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Thanks

FRANCESCO PUGLIESE | francesco.pugliese@istat.it